Tutorial Lab 12.6 (Formatively Assessed):

The FET Differential Amplifier

Preliminary Matter: Setting up the Cadence Environment

Please read through sections 1 through 3 of the tutorial lab 12.1 instruction booklet/handout before entering these commands into your terminal.

For this course, you will use the Cadence Virtuoso Design Environment (referred to simply as Cadence). Cadence is a Linux based software suite for designing full custom integrated circuits. In order to use Cadence you will first need to set-up the Cadence environment.

5. Open a terminal window and ensure that you are in your home directory using the following command:

```
cd ~
```

Note: 'cd' stands for change directory. The solitary tilde symbol (~) is a shortcut/alias for your home directory (/home/username).

6. Under your home directory, use the following commands to load the Cadence environment:

```
module use /eda/cadence/modules
module load course/EENGM6030
```

Note: The two 'module' commands load the required shell/Cadence environmental variables for this course (paths, names, aliases…) In particular, this includes a reference to the process design kit (PDK) used in this course, the GPDK180. You will need to do this every time you open a terminal, since the sourced environment only applies to the current terminal (until it's closed).

7. The AICD folder will be used for all your Cadence related activities for this course. **You should always make sure you are in your AICD folder before you launch Cadence.** To run cadence, use the following commands:

```
cd ~/AICD
virtuoso &
```

.

1. Learning Objectives of this Lab:

From the previous lab, you should:

- Understand the concept of channel length modulation and the dependency of the MOSFET output impedance on channel length.

- Understand the concept of and rationale for cascoding. Apply cascoding to voltage amplifiers and current mirrors in order to increase output impedance and hence reduce the dependency of output current on output voltage.

- Understand the need for a symbol cellview. Generate the symbol cellview for a schematic. Create and simulate a testbench for a user generated symbol. Descend into and out of a symbol cellview.

In this lab, you will:

- Create, and simulate in terms of DC-sweep, a CMOS differential voltage amplifier with resistive and current source loads.

- Understand the need for feedback to set the DC operating point of a fully differential amplifier.

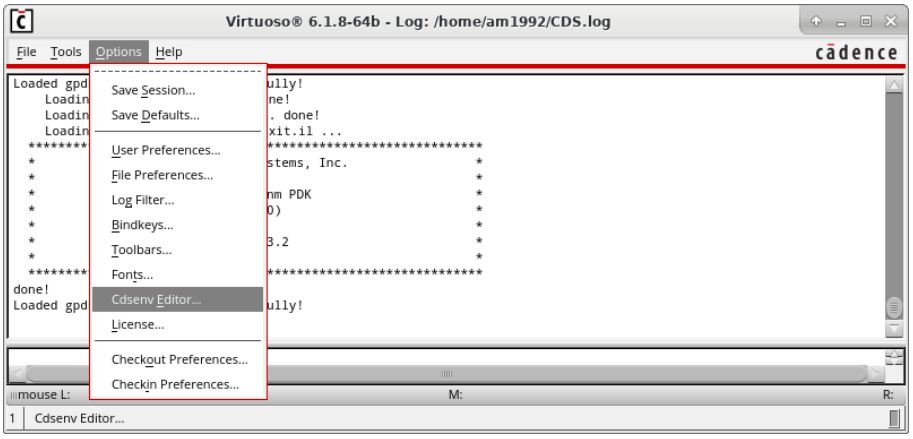
- Create, and simulate, a CMOS FET differential-to-single-ended transconductance amplifier (TCA) with a current mirror load and PFET input devices.

2. Cadence Documentation, Help, and Licensing

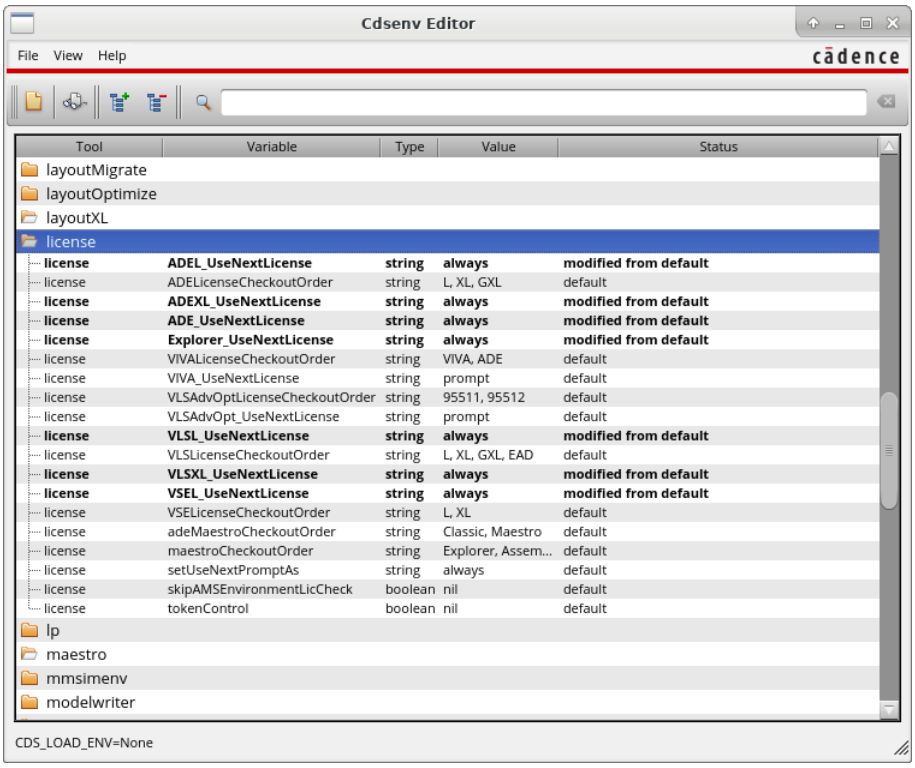
Local and online help is available from the help menus (**Menu: Help → User Guide** or by pressing **F1**), or by typing 'cdnshelp &' at the command prompt (assuming you have setup your terminal environment).

PLEASE NOTE: WHEN LAUNCHING A TOOL IN THE CADENCE DESIGN ENVIRONMENT YOU MAY SOMETIMES GET A WARNING ABOUT A LICENSE BEING UNAVAILABLE FOR THE TOOL, AND REQUESTING PERMISSION TO OBTAIN A HIGHER-TIERED LICENSE. YOU SHOULD SELECT 'ALWAYS' TO THESE MESSAGES.

If you accidently deny a license request, or you cannot access a license, you will need to modify the Cadence environment licence/permissions from the CIW. From the CIW menu go to **Menu: Options → Cdsenv Editor**. Once in the environment editor, go to the license folder, and make sure all licenses (in bold) are set to '**always'** – if any entries are listed as '**never**' you will need to manually change/retype them to '**always**'.



< Figure X.1. The Cadence Environment Menu Option from the CIW Window. >



< Figure X.2. The Cadence Environment Editor. Note, all licenses should be set to 'always'.>

3. Getting Started with Cadence:

For a refresher/introduction on how to start a Cadence session please read through sections 1 through 3 of the tutorial lab 12.1 instruction booklet/handout.

3.1 Dealing with Lock files

PLEASE NOTE: IF A SCHEMATIC ISN'T PROPERLY CLOSED DOWN (FOR EXAMPLE, DUE TO A DROPPED REMOTE CONNECTION), THE SCHEMATIC MIGHT BECOME EDIT LOCKED. IF THIS HAPPENS YOU WILL NOT BE ABLE TO OPEN THE FILE FOR EDITING. YOU WILL ONLY BE ABLE TO OPEN THE FILE FOR READING UNTIL YOU REMOVE ANY/ALL LOCK FILES.

There are two ways you can unlock files:

(4) The first is to manually delete any edit lock files. To do this, go to your '~/AICD/Designs' folder and find the folder representing the schematic e.g., for lab1 this would be '~/AICD/Designs/lab121_vdivider/schematic' if this is what you called your schematic. Delete all the .cdslck lock files in this folder and you should be able to open your schematic for editing.

(5) You can also automatically release edit locks with the command:

```
clsAdminTool -are .
```

To use this command, you have to be in your AICD folder (the full-stop is shorthand for the current directory path). If you are not in your AICD folder, you have to specify the path (relative or absolute) to your AICD directory.

Since this is a Cadence command, you have to ensure that you have set up the terminal environment for Cadence (in particular, the modules must be loaded as in step 2, 'Preliminary Matter'). It is best to use the terminal window that you used to start Cadence, since the environment was already set up. Note that you will not be able enter any commands into your terminal if you did not start cadence with the '&' character. By example, to release any lock files after opening a new terminal, the full command set would be,

```
cd~
module use /eda/cadence/modules
module load course/EENGM6030_2021
clsAdminTool -are AICD
```

For more information on this command, use

```
clsAdminTool -help
```

4. Circuit Simulation Exercises: Part 1

With guidance, you will:

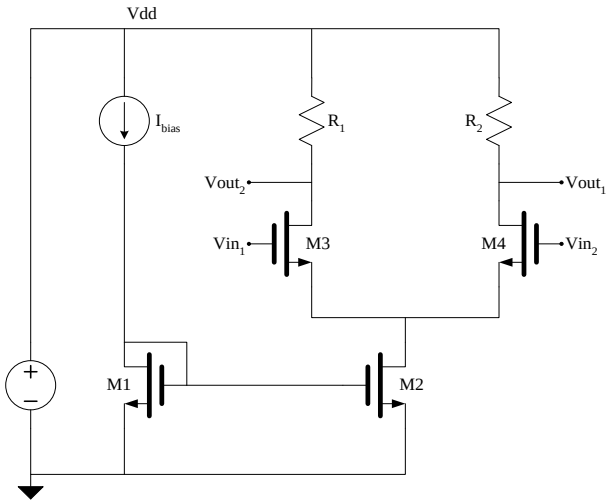
- Create, and simulate in terms of DC-sweep, a CMOS differential voltage amplifier with resistive and current source loads.

Differential amplifiers are fundamental components in Analogue IC design. At their simplest, these circuits amplify the difference between their inputs (the differential mode of operation) and reject/supress what is common between their inputs (the common mode of operation). Hence, interfering signals – internal or external – can be eliminated where they are common to both inputs of the amplifier. This gives differential amplifiers an advantage in sensing and test-and-measurement applications.

Differential amplifiers are most commonly differential-input/differential-output or differential-input/single-ended-output. Single-ended-input/differential-output amplifiers do exist but are usually conFigured from fully differential amplifiers by grounding one of the inputs. For a more detailed overview of differential amplifiers refer to sections 4.4 (Balanced-Output, Balanced Input Amplifiers) and 4.5 (Single-Ended Output Amplifiers and Operational Amplifiers) of the background material.

4.1 The Basic Resistor Loaded Balanced Amplifier

(4.1.1) Create the schematic for the basic balanced amplifier circuit in Figure 1. A summary of the circuit's operation is given in section 4.4 of the background material.

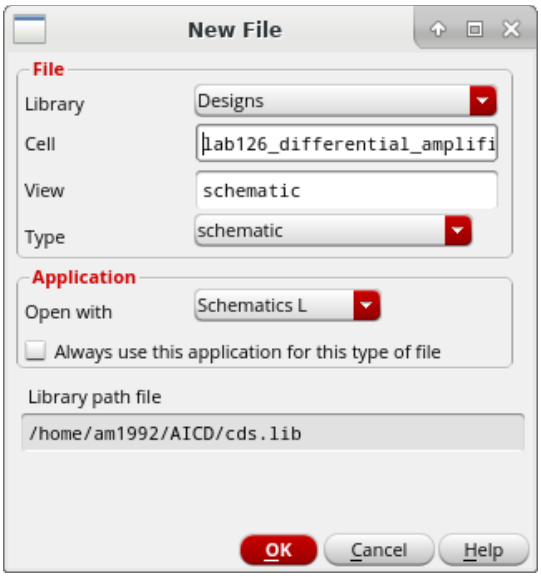


< Figure 168. The Basic Resistor Loaded Balanced Amplifier.
Reproduced from Figure 21, section 4.4 of the background material. >

(4.1.2) From the Library Manager window, select the 'Designs' library and choose **Menu: Files à New à Cell View** to invoke the New File form (Figure 2).

In the 'Cell' text field, choose a meaningful cell name (e.g., lab126_differential_amplifier). In the 'View' text field, type 'schematic' as the view name and from the 'Type' drop down menu choose 'schematic' as the view type. In the 'Application' section of the dialog box, choose 'Schematics L' from the 'Open with' drop down menu. OK the form.

(4.1.3) You should be presented with a blank **Virtuoso Schematic Editor L** schematic entry window for the new cellview.



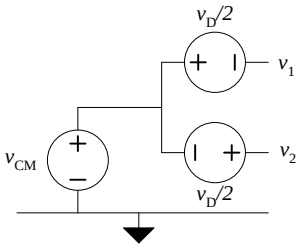
< Figure 169. The New File Form >

(4.1.4) In your schematic editor window, from the GPDK180 library, instantiate and place four **nmoshv** NMOS FETs (two for the input pair, and two for the tail bias current mirror).

*Recall, there are several ways of rotating/flipping a part. When adding an instance to your schematic you can rotate and/or flip the item before you place it on the canvas using the **Rotate**, **Sideways**, and **Upside-Down** buttons on the Add Instance form. Alternatively, you can rotate/flip the components after placing them by selecting the component and using the **rotate** or **flip** commands from the menu/toolbar. You can also flip or rotate a component by using **F3** with the **move/stretch** commands. In general, pressing F3 while a command is in progress gives you a more detailed set of options for that command. When moving an item using the **Move** or **Stretch** commands, pressing F3 will bring up a dialog box with more detailed options including **Rotate**, **Sideways**, and **Upside-Down** buttons.*

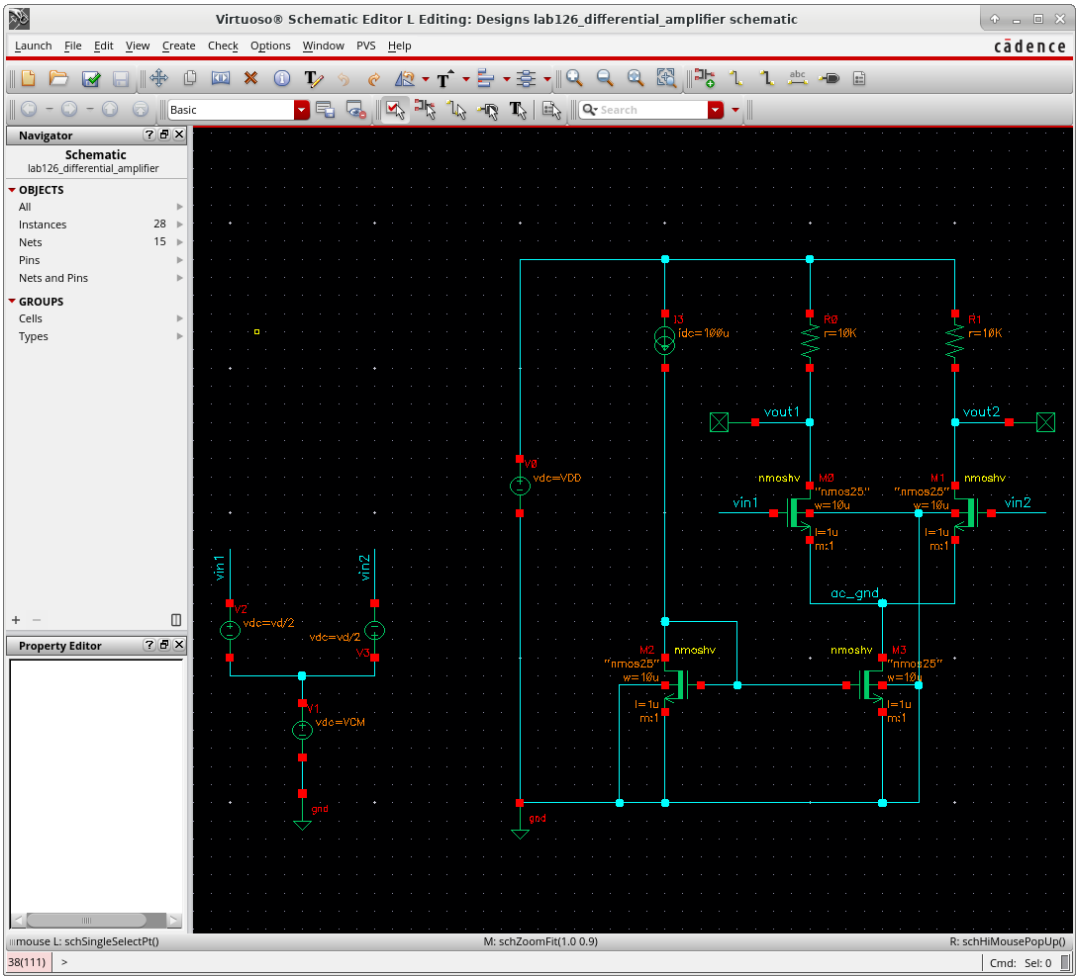
(4.1.5) Instantiate and place an ideal **vdc** source for the supply voltage, an ideal **idc** current source for the tail current reference, and two ideal resistors (**res**) for the balanced loads. All ideal components are from the analogLib library.

(4.1.6) Finally, to drive the circuit, use a single **vdc** for the common-mode input voltage (VCM) and add two split **vdc** inputs for to the differential-mode input voltage (v_D). This will enforce the common-mode voltage (CMV) principle rather than having separate VCMs driven by the same parameter for the two inputs. **Note the opposing orientation of the two v_D/2 vdc supplies – this gives a total voltage of v_D.** See Figure 3 (4).



< Figure 170. Common-mode voltage driver circuit. $v_D = (v_1 + v_2)$. >

(4.1.7) Parameterise your design. Set the tail current reference to 100uA (**DC Current** = 100u) For the DC supply voltage, set the voltage attribute to the string '*VDD*' (**DC Voltage** = *VDD*). For the input set the common-mode voltage component to the string '*VCM*' (**DC Voltage** = *VCM*), and the two differential-mode components to the string '*vd/2*' (**DC Voltage** = vd*/2*). Set both load resistors to 10KΩ (**Resistance** = 10K). Dimension all NMOS devices 10um/1um (**width** = 10u, **length** = 1u). Leave all other attributes unchanged.



< Figure 171. The Resistor-Loaded Differential Amplifier Schematic as in Figure 1>

(4.1.8) Wire the components as in Figure 4 using the **Wire (narrow)** command.

(4.1.9) Using the **Create Wire Name** command: name the two input nets vin1 and vin2 (the gates of the differential input pair), name the two output nets vout1 and vout2 (the drains of the differential input pair), and name the tail bias point ac_gnd (the drain of the tail device / the common sources of the differential input pair). Use the **noConn** connector from the **basic** library on the outputs to avoid floating-net warnings. Keep the schematic neat.

(4.1.10) To connect the inputs to the amplifier to the driver circuit, we will create an implicit connection using wire names. Name the outputs of the common-mode driver circuit vin1 and vin2. This will create a connection between the vin1 of the driver circuit and vin1 of the amplifier, and a connection between vin2 of the driver and vin2 of the amplifier. See Figure 4.

REMEBER: FOR AN NMOS DEVICE, YOU MUST ALWAYS CONNECT THE BODY/BULK TERMINAL TO THE MOST NEGATIVE NET (WIRE) OF THE CIRCUIT. IN FIGURE 4, THIS IS THE GROUND TERMINAL. FOR A PMOS DEVICE (NOT SHOWN) THE BODY/BULK TERMINAL MUST ALWAYS BE CONNECTED TO THE MOST POSITIVE NET (WIRE) OF THE CIRCUIT, USUALLY THE SUPPLY VDD. THIS IS NECESSARY TO ENSURE THAT THE CHANNEL WILL BE PROPERLY ESTABLISHED.

(4.1.11) Check and save your schematic using the **Check and Save** command. Remember to use the output area of the CIW to diagnose any problems with your schematic and/or simulations if there are any warning or error messages when checking, saving, or simulating your work.

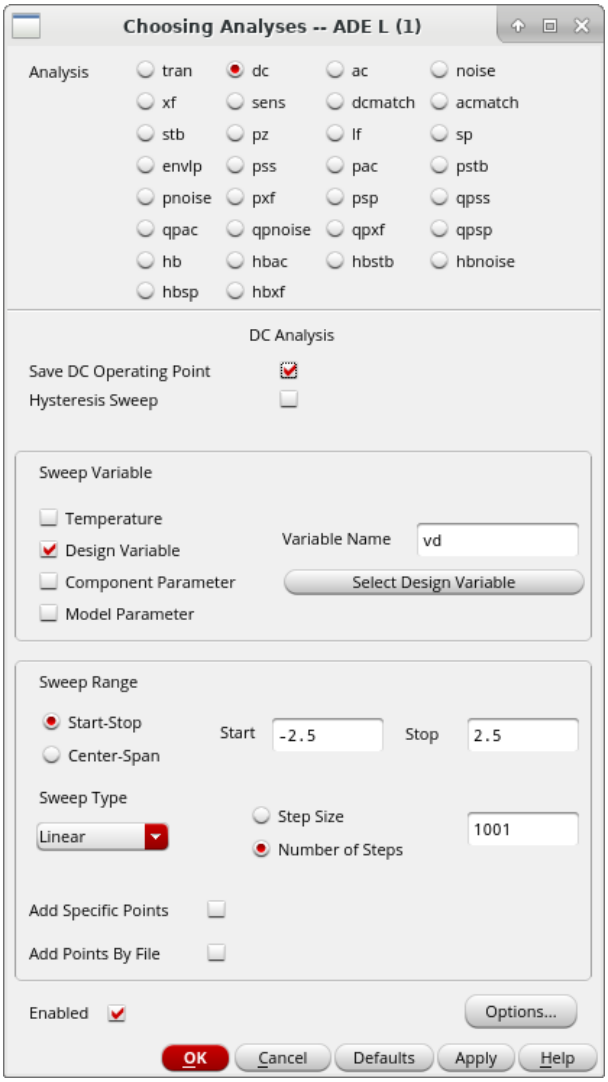
4.2 Simulating the Basic Differential Amplifier

(4.2.1) We will simulate the amplifier using a DC Sweep Analysis. From the Schematic Editor window, launch the Analogue Design Environment (ADE) – **Menu: Launch à ADE L**.

(4.2.2) Copy the design variables from the schematic into the ADE. For this circuit, these are the DC voltages from the four **vdc** components – 'VDD', 'VCM', and 'vd'. From the ADE window, choose **Menu: Variables à Copy From Cellview**. You should now see the VDD, VCM, and vd variables in the 'Design Variables' list of the ADE. Initialise VDD *= 2.5V*, VCM *= 1.25V,* and vd = 0V. These values are used to set the DC operating point of the amplifier circuit.

(4.2.3) We want to plot the differential output voltage (vout1 and vout2) and the AC ground point (ac_gnd) as we sweep vd from -2.5 to 2.5V. From the ADE window, go to **Menu: Analyses à Choose…** or use the '**Choose Analyses**' button on the right-side of the ADE window (top button). The 'Choosing Analyses – ADE L' form will open.

In the choosing analyses form: Select the '**dc**' option button. Check the 'Save DC Operating Point' box. Check the 'Design Variable' box as the sweep variable. Type vd as the variable name. Choose the 'Start-Stop' option button with Start = -2.5 and Stop = 2.5 for the sweep range. Select 'Linear' from the drop-down menu as the sweep type. Choose the 'Number of Steps' option button with 1001 steps in the sweep. Ensure that the analysis is enabled by checking the 'Enabled' box near the bottom of the form. See Figure 5. OK the form. The **dc** analysis should now appear in the 'Analyses' list of the ADE window.

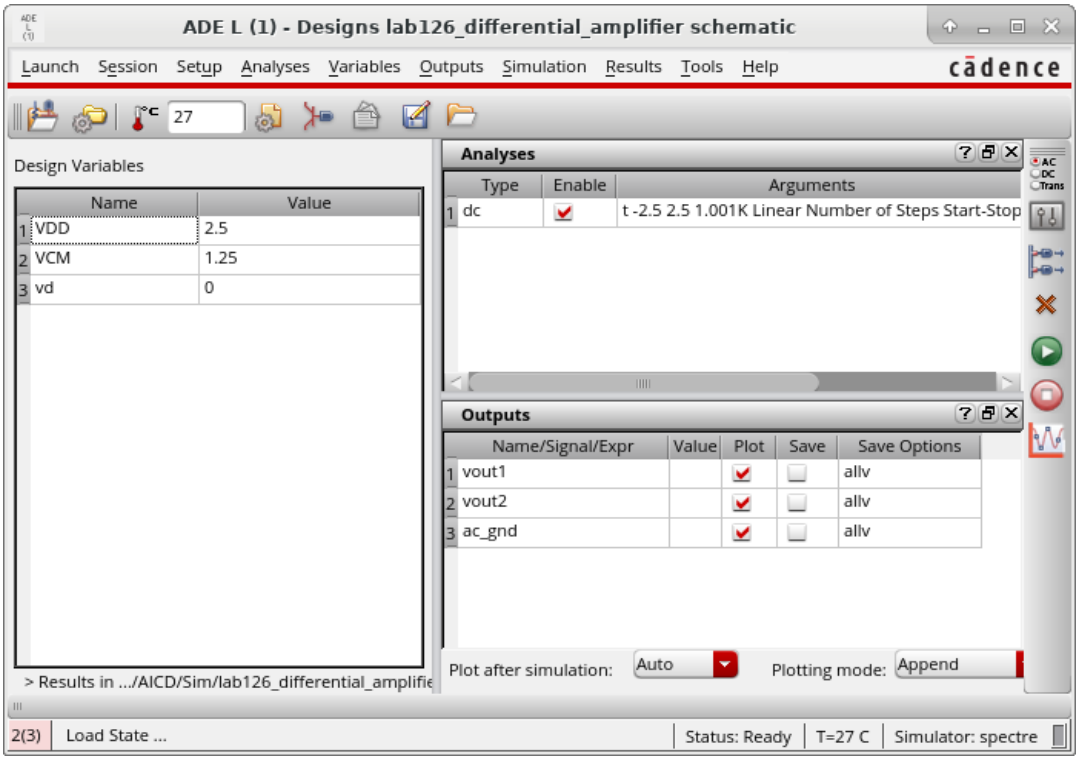


< Figure 172. The Choosing Analyses Form. >

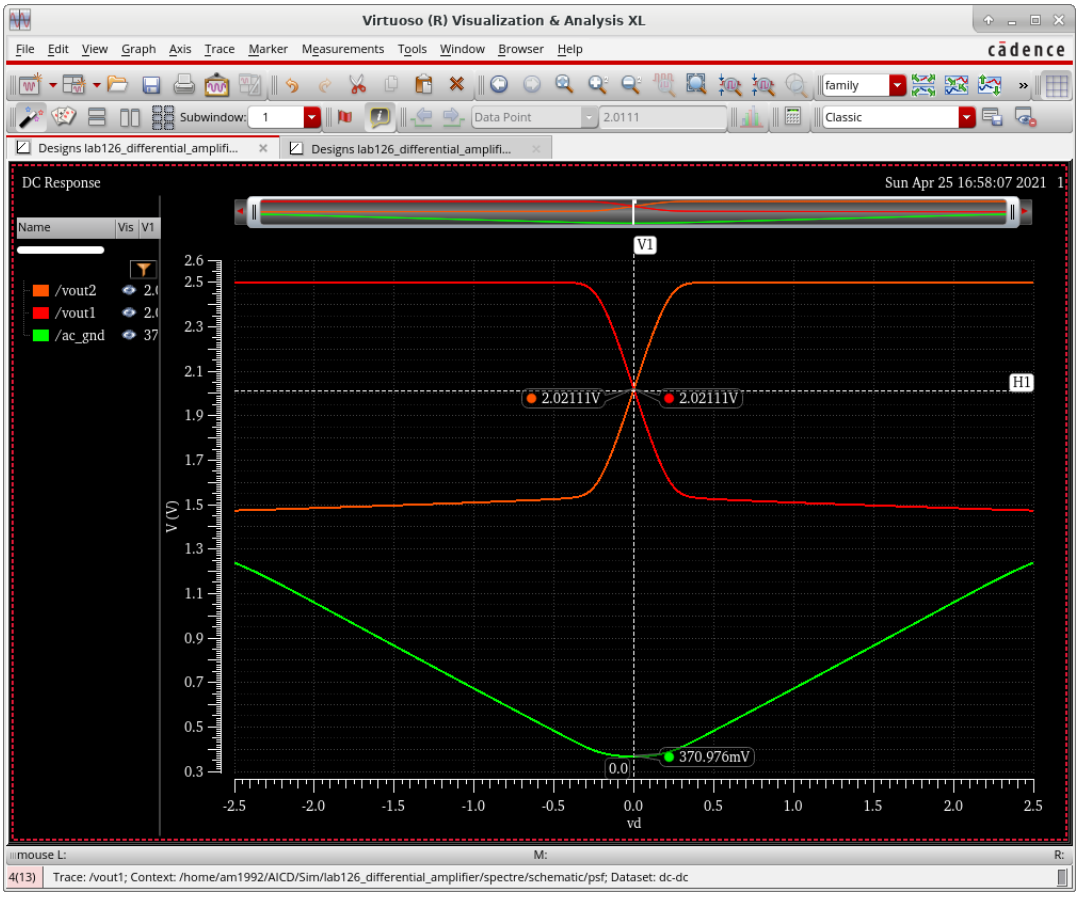
(4.2.4) Select the two output voltages (vout1 and vout2) and the AC ground point (ac_gnd) for plotting. In the Analog Design Environment window, select **Menu: Outputs à To Be Plotted à Select On Schematic** to enter the plotting mode of the ADE**. Press the 'ESC' key to exit the plotting mode of the ADE**.

Recall, if you want to plot a particular voltage, you click on the net/wire. If you want to save and plot the currents, you have to click a terminal ***node*** *of a component (not the net/wire).*

(4.2.5) In the Analog Design Environment window, run the '**Netlist and Run**' command by clicking on the green play button. Both the text output window as well as the waveform output window should open automatically. See Figures 7 and 8.

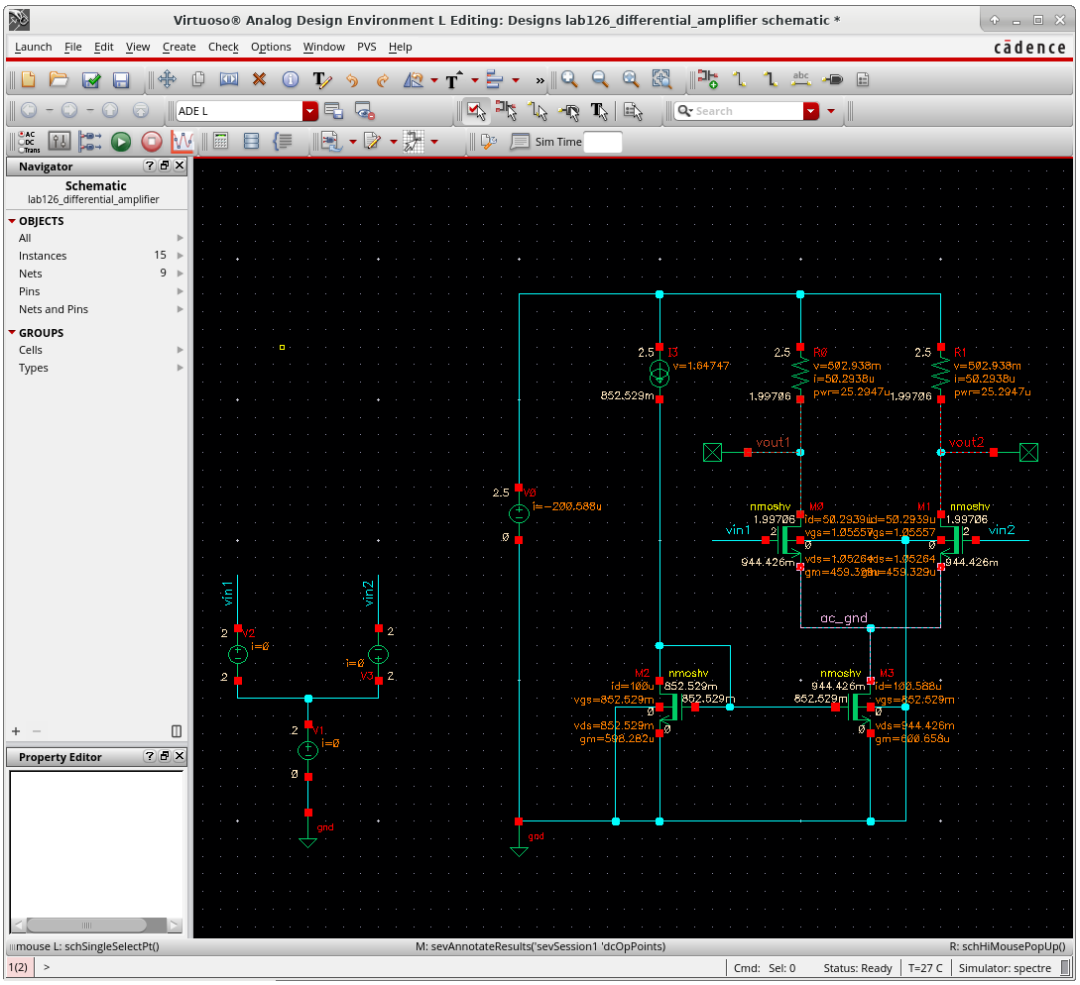


< Figure 173. The ADE with the Design Variables, Analyses and Outputs. >



< Figure 174. Resistor-Loaded Differential Amplifier Output Voltages (vout1, vout2) and the AC Ground Point (ac_gnd). VCM=1.25. VCM=1.25V. VD=-2.5V to +2.5V >

(4.2.6) You can also annotate the schematic, **Menu: Results à Annotate à DC node voltages…** and **Menu: Results à Annotate à DC operating points…** See Figure 6.



< Figure 175 (out of Figure numbering order). Annotated. Resistor-Loaded Differential Amplifier Schematic, this is actually showing the node voltages for a VCM = 2V as used below>

(4.2.7) **Discussion**. Referring to the plot of Figure 8. The differential output (vout1, vout2) sits on top of the common mode output voltage (V_{CMO}) – this is the DC operating point when the input voltage (vd) is zero.

When **vd=0V**, the tail current (100uA) is split equally between the two branches of the differential amplifier (50uA through each). In this case, the common-mode output voltage is 2.0111V (indicated by the vertical marker at the *vout1/vout2* cross point). This can be calculated from,

$$V_{OUT} = V_{DD} - I_{DS}R_L = 2.5 - \frac{1}{2}(100\mu A)10k\Omega = 2V$$

Where I_{DS} is the current in the tail device (the NFET that is biasing the differential amplifier, M3). In these equations we assume that is the same as the circuit bias current for simplicity.

When **vd > |0.3V|**, the tail current is steered almost entirely into one side of the differential amplifier. At one extreme, the maximum vout is defined by zero current through the load, in which case

$$V_{OUT} = V_{DD} - I_{DS}R_L = 2.5 - (0)10k\Omega = 2.5V$$

At the other extreme, the minimum vout is defined (mostly) by the maximum voltage drop across load when all of the bias current is drawn through the load, in which case

$$V_{OUT} = V_{DD} - I_{DS}R_L = V_{DD} - (100uA)10k\Omega = 1.5V.$$

In analogue design, it is the transitory region around vd=0V in which we are interested. This is the analogue region of operation. In this region, when **vd < |0.3V|**, we see the typical input/output voltage transfer curve that we have previously studied with the single-ended amplifier designs. However, in this case, we have two complementary outputs – the inverting output and the non-inverting output. (Which output is inverting/non-inverting depends on which input you are comparing it to – the output on the same side as the input is inverted with respect to that input. The output on the opposite side to the input is non-inverting with respect to that input.)

The slope of the transfer curve defines the gain of the amplifier. As with the single-ended amplifier design, the differential gain is given by the product of the transconductance (g_m) of the input devices and the load resistance,

$$A_V = 2g_mR_L$$

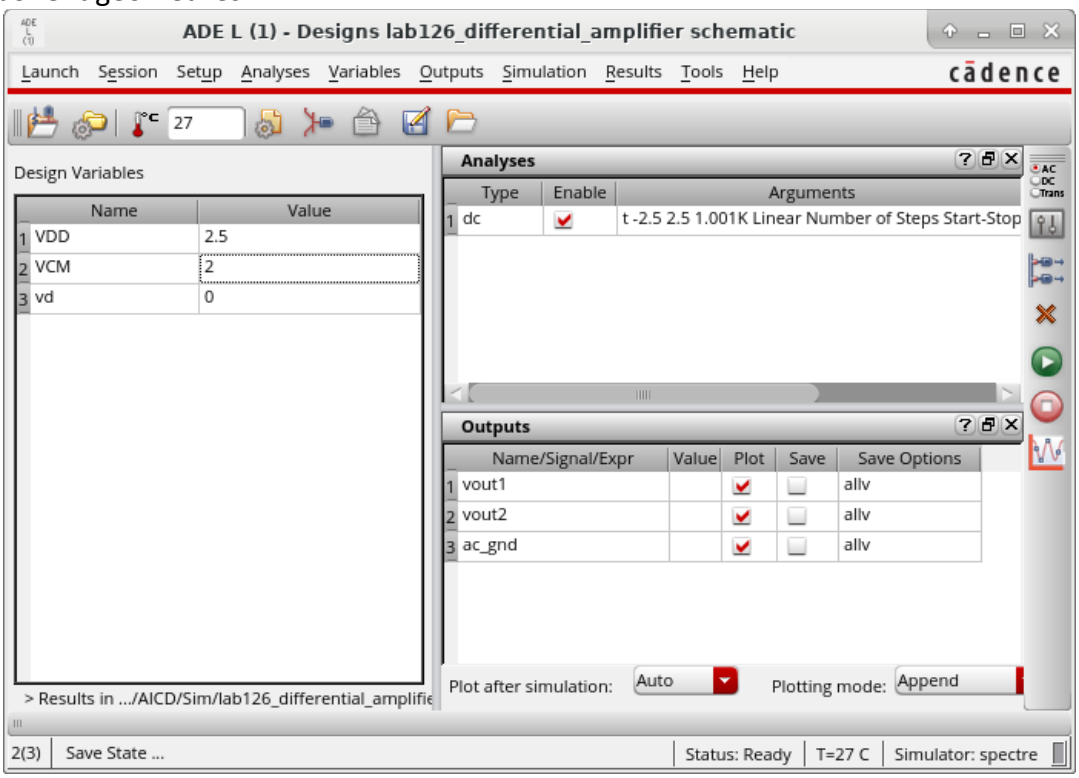
Note that the transconductance is halved at vd=0 because the tail current is shared between the input devices.

The output of the differential amplifier may be taken single-ended or differentially. A single-ended output may be attained by taking one output with respect to ground. A balanced or differential output may be attained by taking the difference between the two outputs. The balanced output is double the gain of the unbalanced gain – hence the '2' multiplier in the gain equation above. Later in this lab, we will use the calculator to plot the differential voltage transfer curve, from which the gain can be calculated.

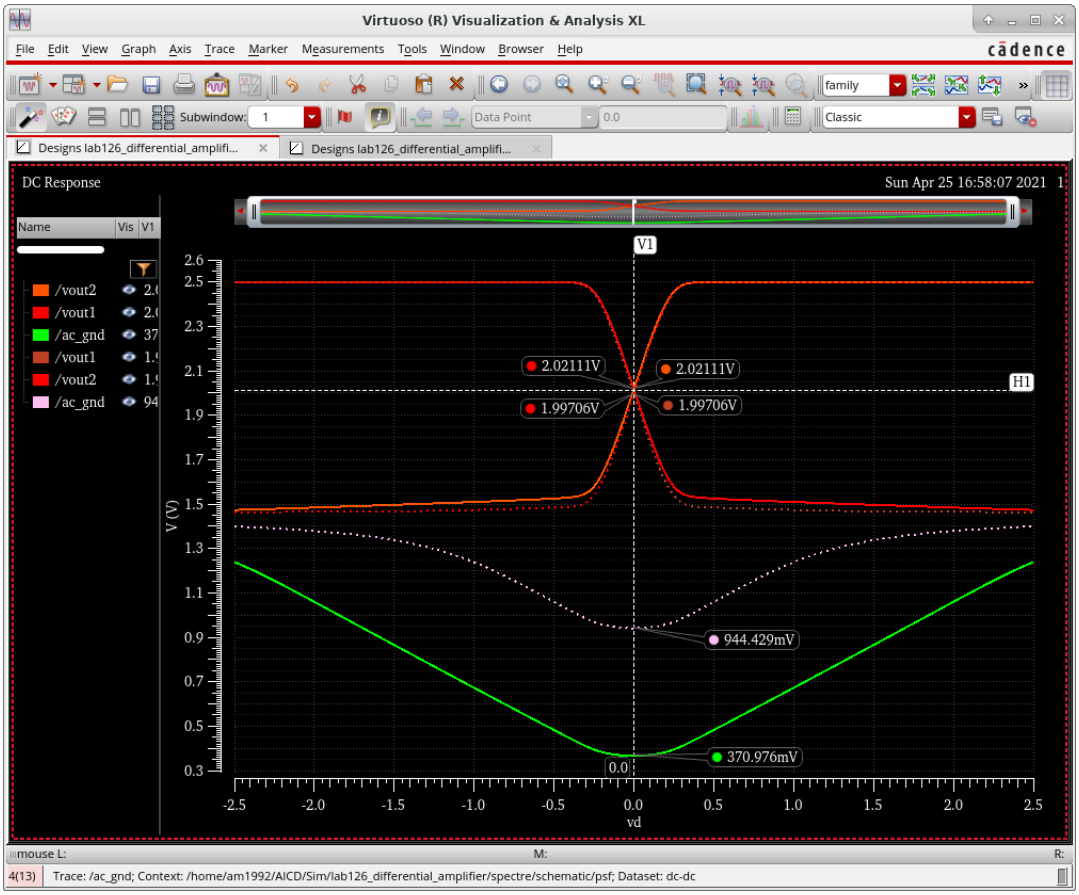
One final point to note, the common-source AC ground point of the differential amplifier is also plotted (ac_gnd) alongside the output voltages (vout1, vout2). Here, we see that there is some variation of the voltage at this point as vd is swept from -2.5V to 2.5V. This is caused by the source voltage of the dominant input FET carrying the current, 'pulling up' the AC ground point as the voltage on its gate is increased (This is a high impedance node). If the NMOS input pair were ideal linear devices, we would indeed see no variation of the voltage at the AC ground point. However, their linearity is only assumed over small excursions of the input voltage, so as the voltage on the gate of the dominant NMOS FET increases, its source will follow at a v_{gs} below the gate.

Furthermore, the tail bias device has a current that is dependent on its drain voltage – due to channel length modulation. This non-ideal behaviour of the bias device will compound the variation on the AC ground point. We can lower the variation of the AC ground node by increasing the common mode input voltage (VCM). If we re-simulate with a common mode input of VCM=2V (Figures 9 and 10), we can see how this lowers the variation on the AC ground node, with little change in the output voltages (vout1 and vout2). Set the plotting mode of the ADE to append (bottom-right drop-down menu.)

Note that the variation of the AC ground point is not normally a problem and so does not define what the VCM should be. If it is a problem, then the designer will redesign the component geometries.



< Figure 176. The ADE with the Design Variables, Analyses and Outputs. >



< Figure 177. Resistor Loaded Differential Amplifier Output Voltages and AC Ground Point. VCM=1.25V (solid) & VCM=2V (dotted). VCM=1.25V. VD=-2.5V to +2.5V >

5. Circuit Simulation Exercises: Part 2

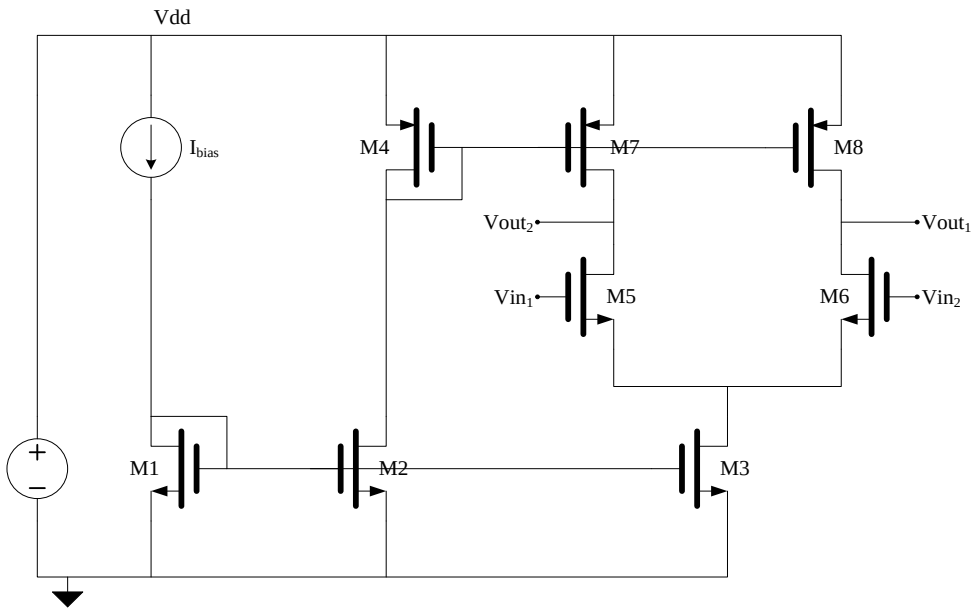
With guidance, you will:

- Create and simulate a current-source loaded (active-loaded) differential amplifier.
- Understand the need for feedback in order to set the DC operating point of a fully differential amplifier.

5.1 The Current-Source Loaded Balanced Amplifier

(5.1.1) In this part of the lab, we will replace the two 10kΩ resistor loads with PMOS current sources biased by an expanded current mirror.

(5.1.2) Modify your existing schematic by constructing the current source loaded amplifier alongside your existing circuit. We will use a proper expansion of the current reference rather than the quick but unrealistic version (with the current source placed in between the two driving FETs, M2 and M4). See Figure 11.



< Figure 178. The PMOS Current-Source Loaded Balanced Amplifier (with Current Mirror Expansion). Reproduced from section 4.4 of the background material. >

(5.1.3) Set all NMOS devices to a W/L ratio of 10um/1um (**Total Width** = 10u, **Length** = 1u). The multipliers of all NMOS devices should be set to one (**Multiplier** = 1).

(5.1.4) Set all PMOS devices to a W/L ratio to 25um/1um (**Total Width** = 25u, **Length** = 1u) to approximately match the transconductances to the NMOS devices (since the NMOS devices have a carrier mobility approximately 2.5 times that of the PMOS devices).

(5.1.5) For the PMOS current mirror/bias device (M4 in Figure 11), set the multiplier to two (**Multiplier** = 2). The multiplier of the PMOS load devices (M7, M8 in Figure 11) should be set to one (**Multiplier** = 1).

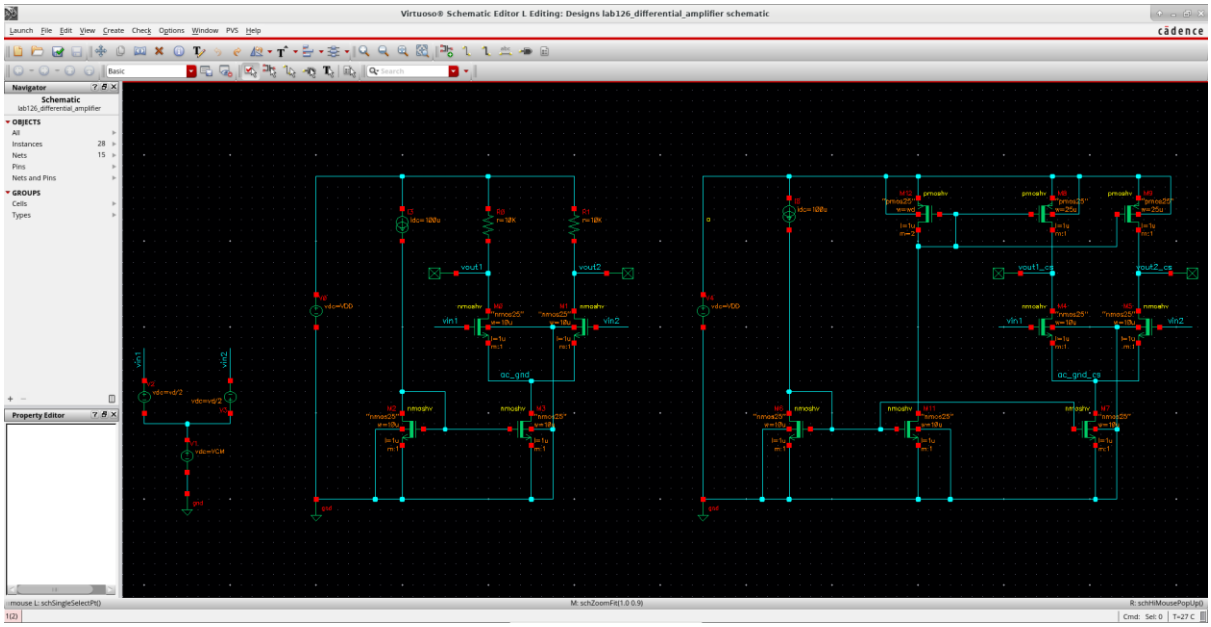
(5.1.6) **Discussion**. Recall, from our discussion in lab 12.2 (paragraph 5.4.13), when biasing the amplifier, we want to set the output DC operating point to approximately mid-rail (VDD/2). For a current-source loaded amplifier, we will only set the output (vout1, vout2) to be mid-rail if the current sourced by the PMOS load devices is the same as the current sunk by the NMOS input devices. That is, we must balance the currents of the NMOS and PMOS devices.

Here, we achieve this balance of the PMOS (source) and NMOS (sink) currents by scaling the PMOS mirror/bias device M4 in fig. 11. Since the tail current is set to 100uA, the current in each branch/side of the differential input pair must be 50uA. We therefore want the PMOS load devices to source 50uA each. We do this by adjusting the PMOS current mirror/bias device (M4) and load devices (M7, M8) to have a ratio of 2:1:1, respectively. This ratio means the PMOS load devices will source half the current that is mirrored into the PMOS mirror device (100uA:50uA:50uA).

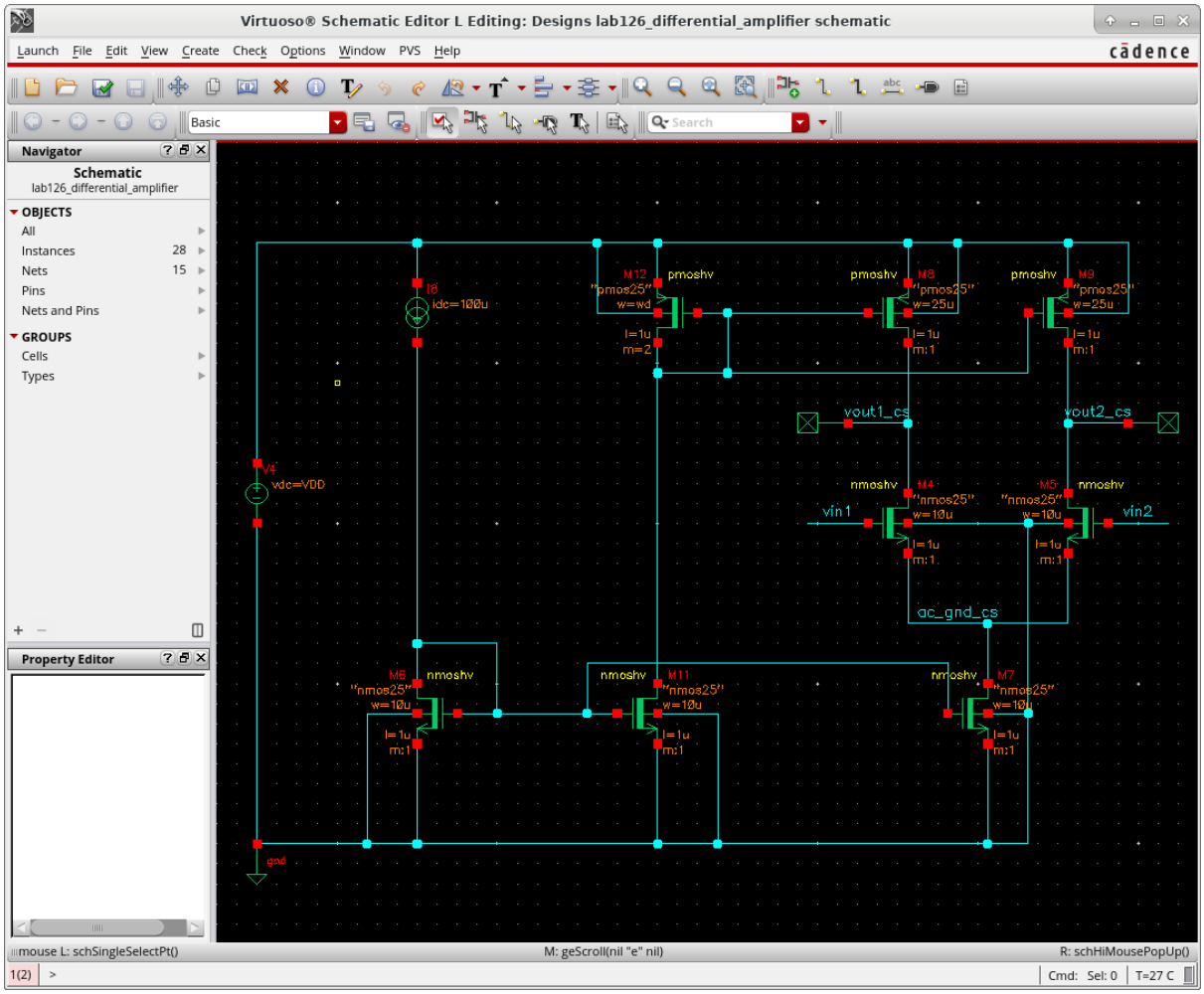
Note: This way of balancing the NMOS/PMOS currents is purely for teaching purposes. In practice this would not work.We would use feedback to drive the PMOS load devices to establish the output DC operating point at approximately mid-rail. See section 5.3.

(5.1.7) Change the name of the output nets to vout1_cs and vout2_cs (this avoids creating an implicit connection between the outputs of the two circuits.) Do the same for the AC ground point – ac_gnd_cs. The inputs can be left as vin1 and vin2 (this allows both amplifiers to be driven by the same driver circuit). See Figures 12 and 13.

(5.1.8) Check and save you schematic using the **Check and Save** command.



< Figure 179. Resistor-Loaded and Current-Source Loaded Differential Amplifiers. >



< Figure 180. Close-Up of the Current-Source Loaded Differential Amplifiers. Note the multiplier values of the PMOS devices from left to right m=2, m=1, m=1. (Ratio 2:1:1). >

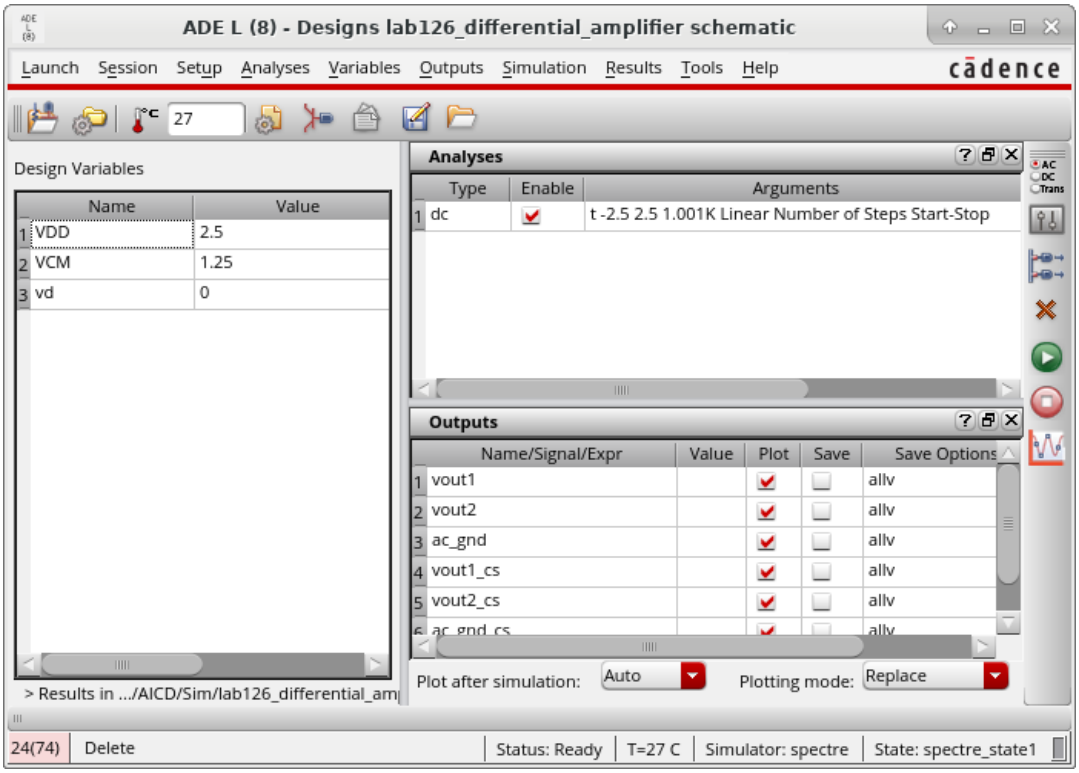
5.2 Simulating the Differential Amplifier using a Swept-DC Analysis

(5.2.1) Return VCM to 1.25V (if you haven't already done so).

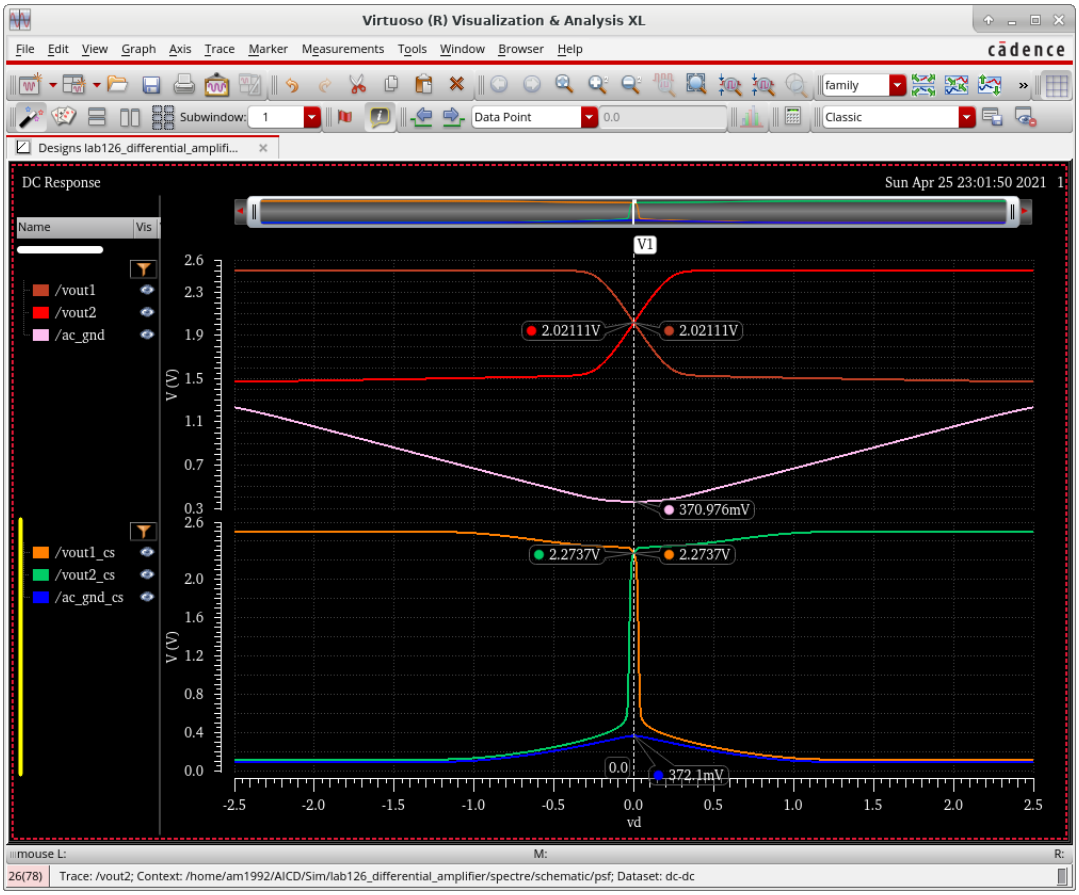
(5.2.2) Select the second amplifier's output nets (vout1_cs, vout2_cs) and AC ground point (ac_gnd_cs) for plotting alongside the first amplifier (vout1, vout2, ac_gnd). **Press the 'ESC' key to exit the plotting mode of the ADE**. See Figure 14.

(5.2.3) Re-run the DC sweep (this should still be set-up form the previous part – if not repeat step 4.2.3). When the simulation has completed, separate the traces for the resistor-loaded and current source loaded differential amplifies. Figures 14 and 15.

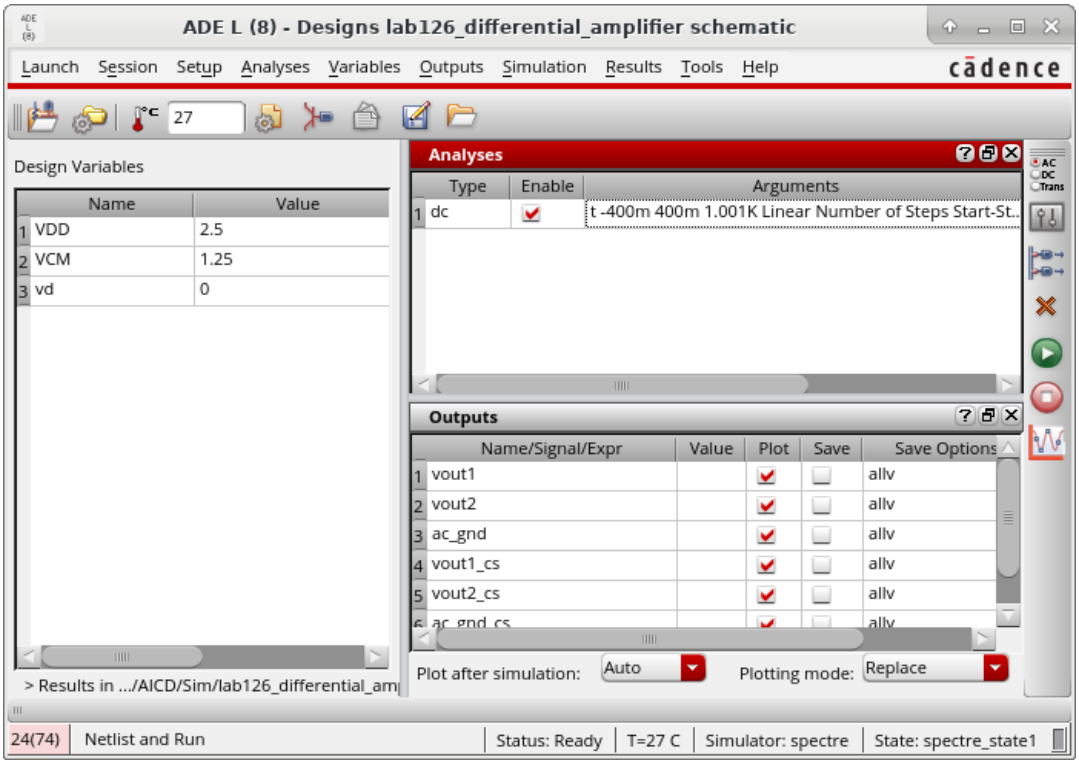
(5.2.4) We are only interested in the transition area around vd=0V – the so-called analogue region of operation. Repeat the simulation again with vd swept between -0.4V and 0.4V for better picture of analogue region of the voltage transfer curve. Figures 16 and 17.



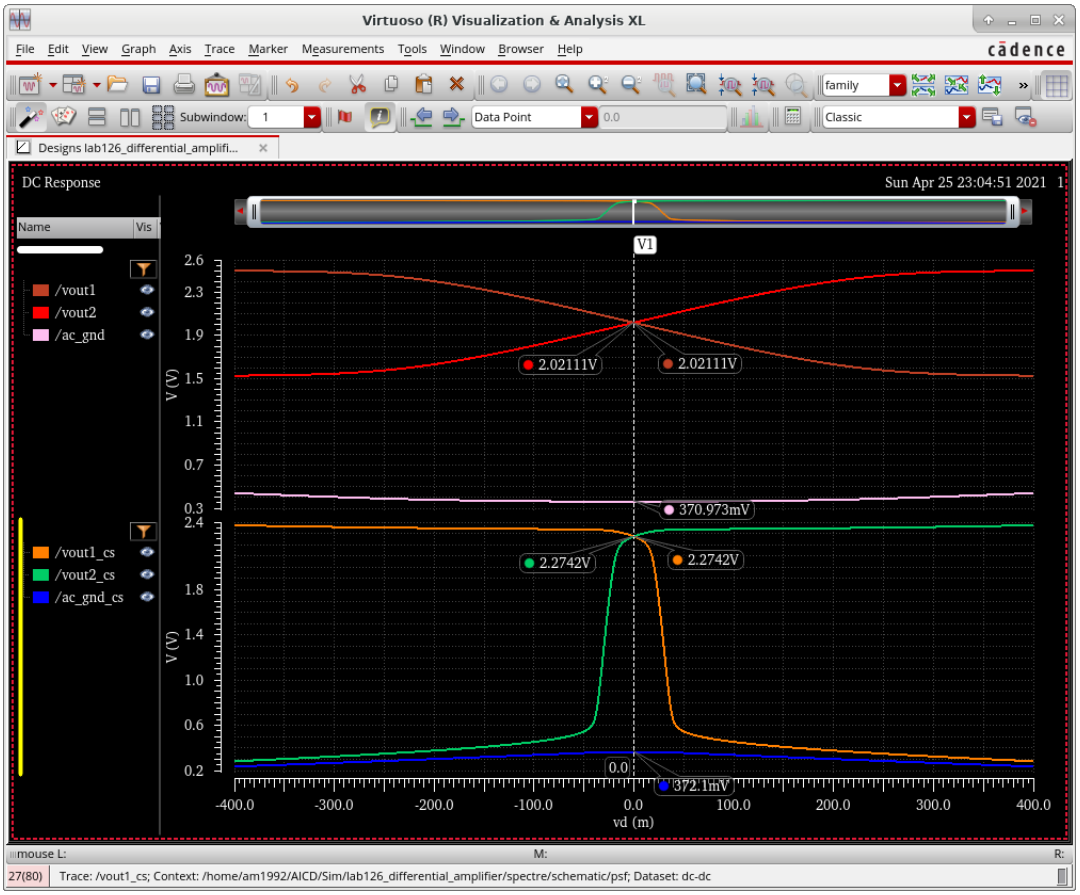
< Figure 181. The ADE with the Design Variables, Analyses and Outputs. >



< Figure 182. Resistor-Loaded and Current-Source Loaded Differential Amplifier Output Voltages. VCM=1.25V. VD=-2.5V to +2.5V >



< Figure 183. The ADE with the Design Variables, Analyses and Outputs. >



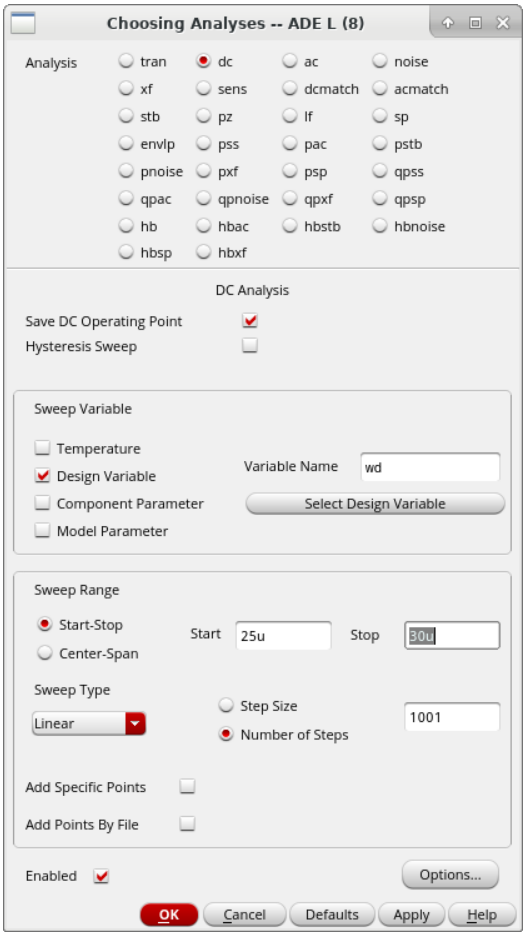
< Figure 184. Resistor-Loaded and Current-Source Loaded Differential Amplifier Output Voltages. VCM=1.25V. VD=-0.4V to +0.4V >

(5.2.5) **Discussion**. Since the gain of an amplifier is given by the slope of its transfer curve, we see that greater load resistance of the PMOS current source gives more gain (as we would expect). However, we can also see there is a significant offset in the output – that is, at vd=0V, we are not operating in the middle of the voltage transfer curve where the gain is at its greatest. In fact, at vd=0V, we are operating very close to the supply rail (2.274V) with no headroom for signal swing and low gain (just where the VTC begins to flatten out).

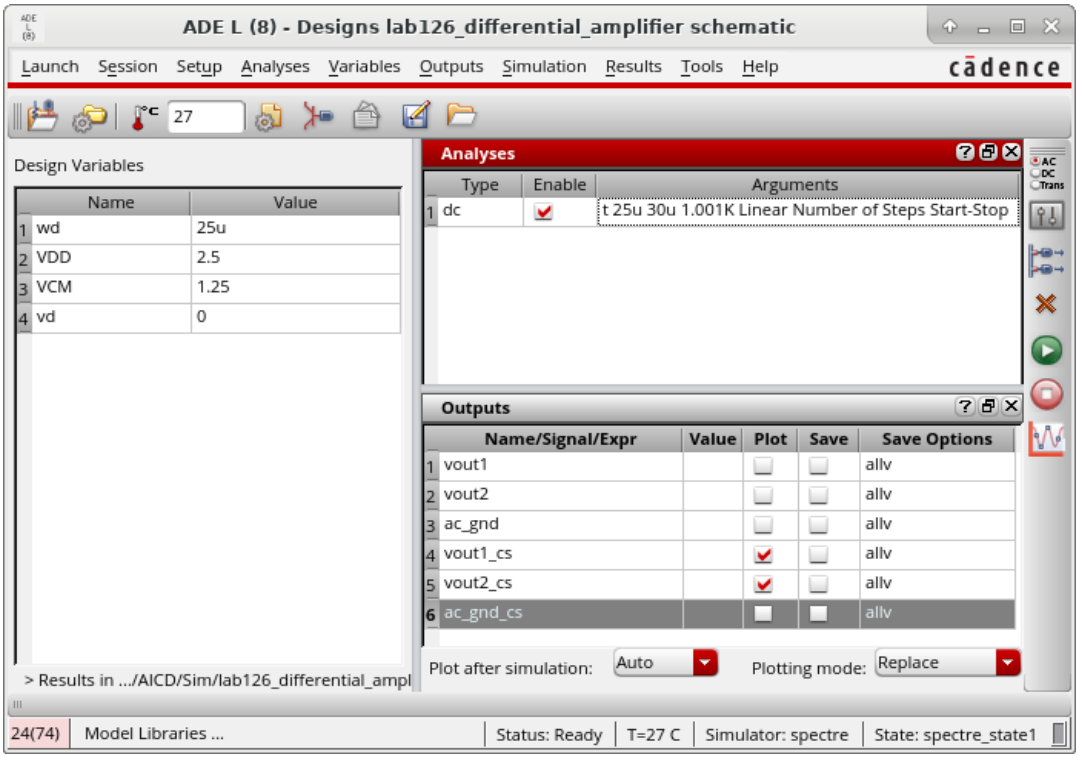
There is an additional problem with design. The vds of the tail device is only 372mV. The device is operating in the triode region. Whilst this isn't a problem for this illustration, in practice the designer would redesign the component geometries and increase the input common mode.

As we have discussed previously, we want to set the output voltage at around mid-rail (VDD/2). In order achieve this we have to precisely match the currents being sourced by the PMOS loads and sunk by the NMOS input devices.

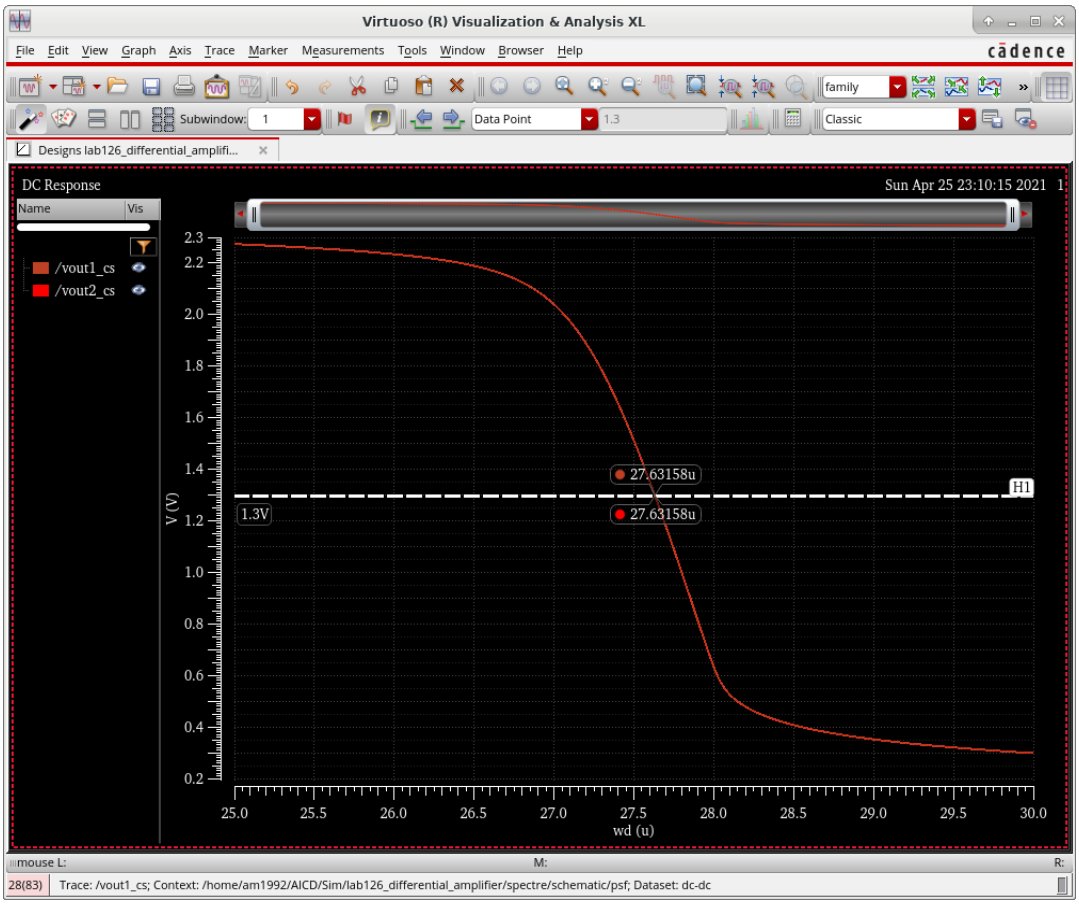
We will run a DC sweep, by parameterising the width of the PMOS mirror/bias device (M4 in fig. 11) to ‘wd’. Modfiy the DC sweep analysis to sweep the width of the PMOS device from 25um to 30um (start-stop, linear, number of steps 1001 - see figs. 18 and 19) and observing when the common mode output voltage (vout1 and vout2 when vd=0V). We see a device width of 27.63um gives a VCMO of 1.3V. See Figure 20.



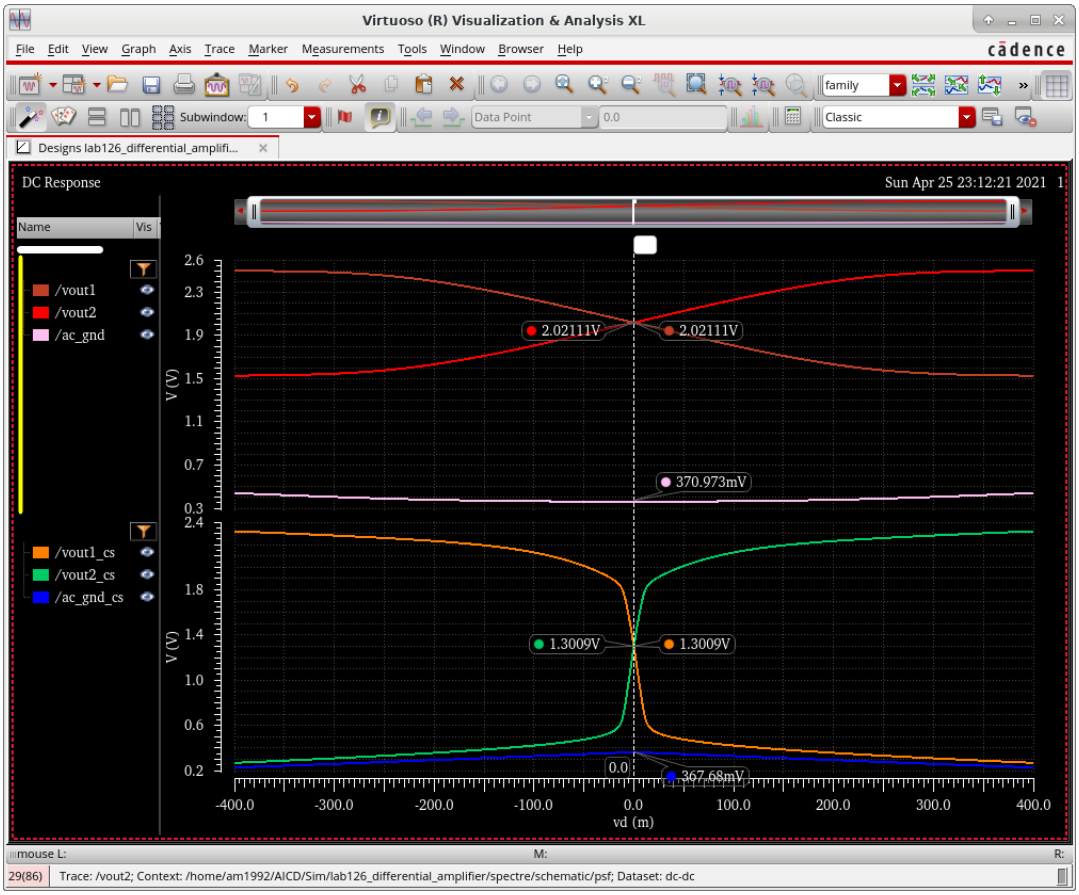
< Figure 185. The Choosing Analyses Form. >



< Figure 186. The ADE with the Design Variables, Analyses and Outputs. >



< Figure 187. Common-Mode Output Voltage of the Current-Source Loaded Differential Amplifier as PMOS mirror/Bias Device is Swept from 25um to 30um. At 27.63um VCMO=1.3V >

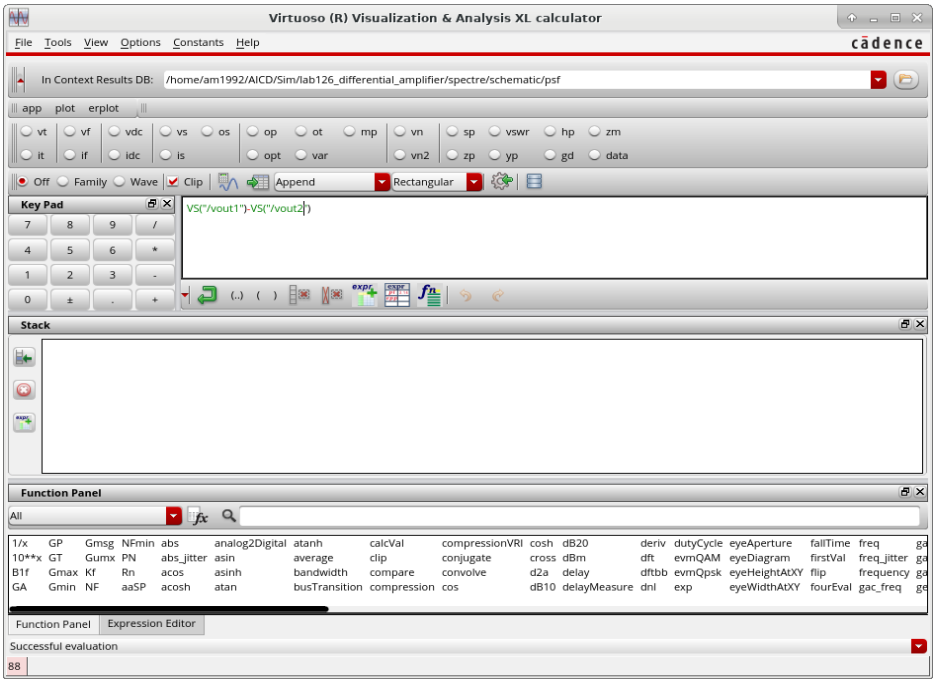


< Figure 188. Resistor-Loaded and Current-Source Loaded Differential Amplifier Output Voltages. VCM=1.25V. VD=-0.4V to +0.4V. PMOS mirror/Bias Device Width = 27.63um >

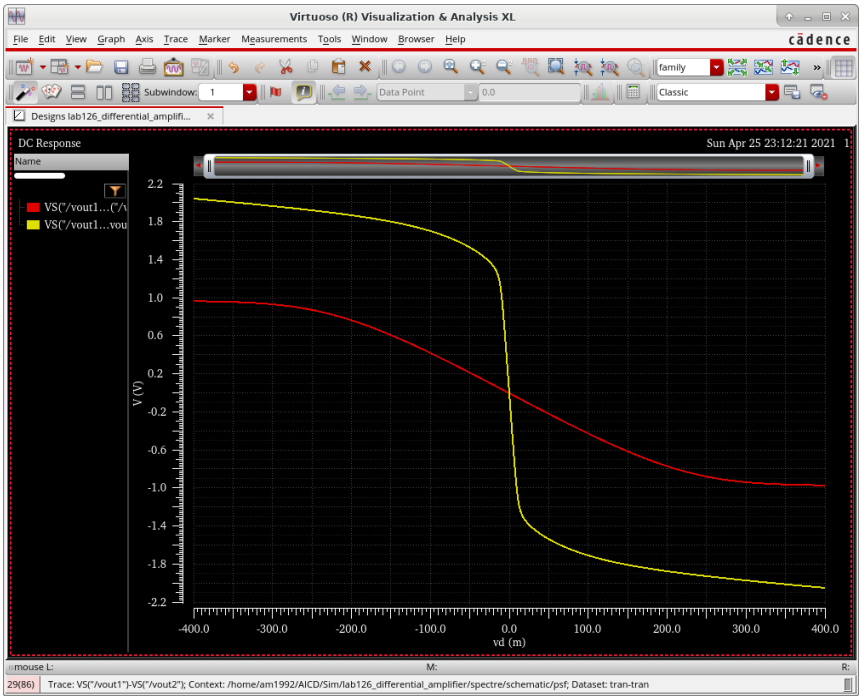
(5.2.6) **Discussion**. Set the PMOS mirror/bias device width the 27.63um and re-run the DC sweep for the differential input voltage vd from -2.5V to +2.5V (see step 4.2.3).

Having precisely adjusted the PMOS mirror/bias device width to 27.63um (length 1um), we see the common-mode output voltage (the DC operating point, at vd=0V) has been centred to approximately mid-rail 1.3009V. **It must be stressed that this method of setting the common-mode output voltage is totally unrealistic in practice. There is simply no way of setting the device widths/lengths with such precision due to variations in process technology. Furthermore, even if we could set the device dimensions with such precision, the circuit would not be able to maintain the DC operating point over a wide temperature range, typically -40°C to 125°C, nor over a varying voltage supply (+/- 10%). Prectical circuits use feedback to establish the common mode output voltage. We will demonstrate this in section 5.3.**

(5.2.7) Use the calculator to plot the difference between the two outputs and note that the gain is increased by 2 (compared to a single output). (If you need a refresher on how to use the calculator, refer to section 4.3 - Using the Built-in Waveform Calculator - of lab 12.3.) In the calculator buffer, construct the expression for the differential gain: VS("/vout1")-VS("/vout2"). Do this for both the resistor-loaded and current source-loaded amplifiers. Plot the VTC differential gain curves. See Figures 22 and 23.



< Figure 189. Calculator Expression for Calculating Differential Output Voltage. >



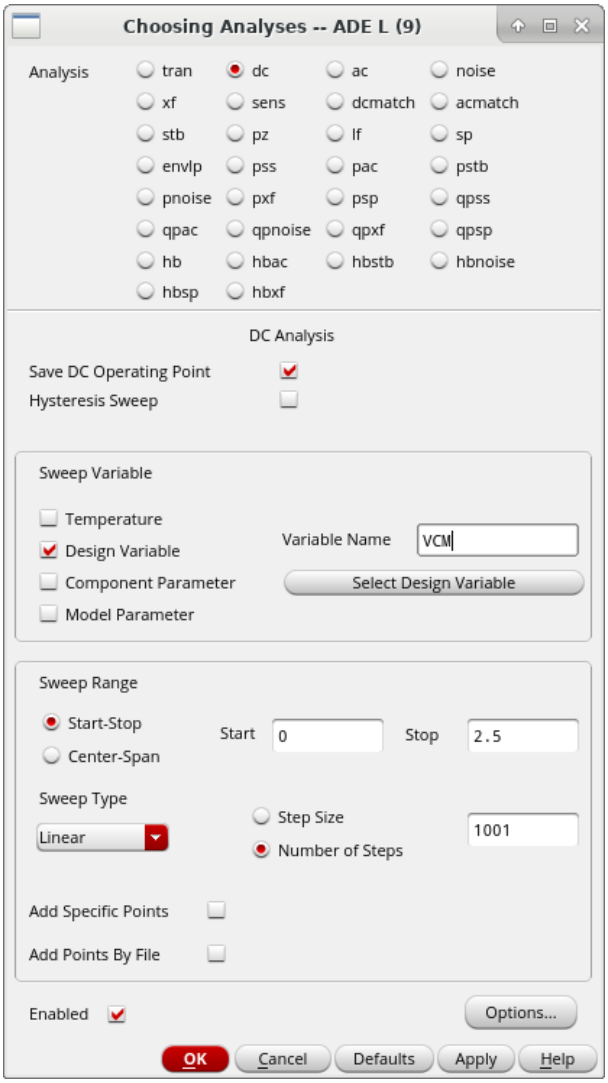
< Figure 190. Differential Output Voltage for Resistor-Loaded (red) and Current-Source Loaded (yellow) Differential Amplifiers. >

(5.2.8) Finally, we want to establish the approximate common-mode output voltage range of the differential amplifiers.

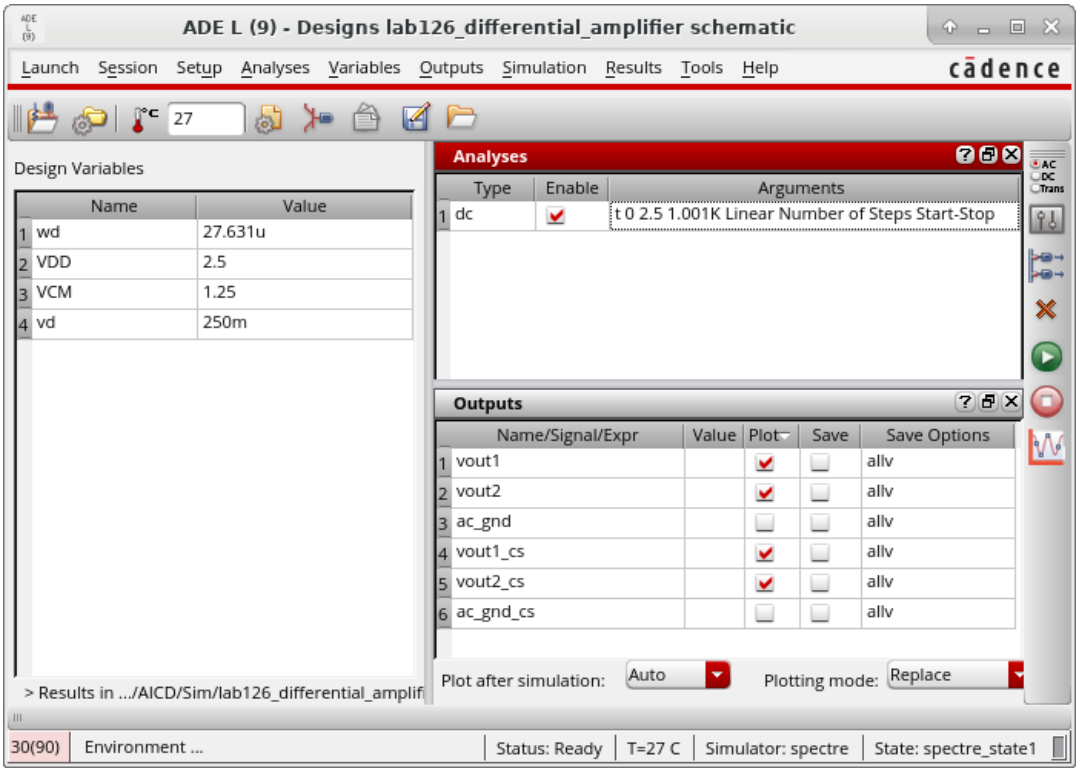
Recall from the introduction: differential amplifiers amplify the difference between their inputs (differential-mode) and reject what is common between their inputs (common-mode). To determine the common-mode output voltage range through simulation, we will apply a fixed differential input voltage of 250mV (vd=0.25V) and sweep the common-mode input voltage (VCM) from 0V to 2.5V. The common-mode range will then be the range over which the common-mode output voltage remains approximately flat at its DC operating point.

In the ADE, set vd = 250 mV. Modify the DC analysis form to make VCM the sweep variable and the 'Start-Stop' option to Start = 0 and Stop = 2.5 for the sweep range. Leave all other properties unchanged. See Figure 24. OK the form.

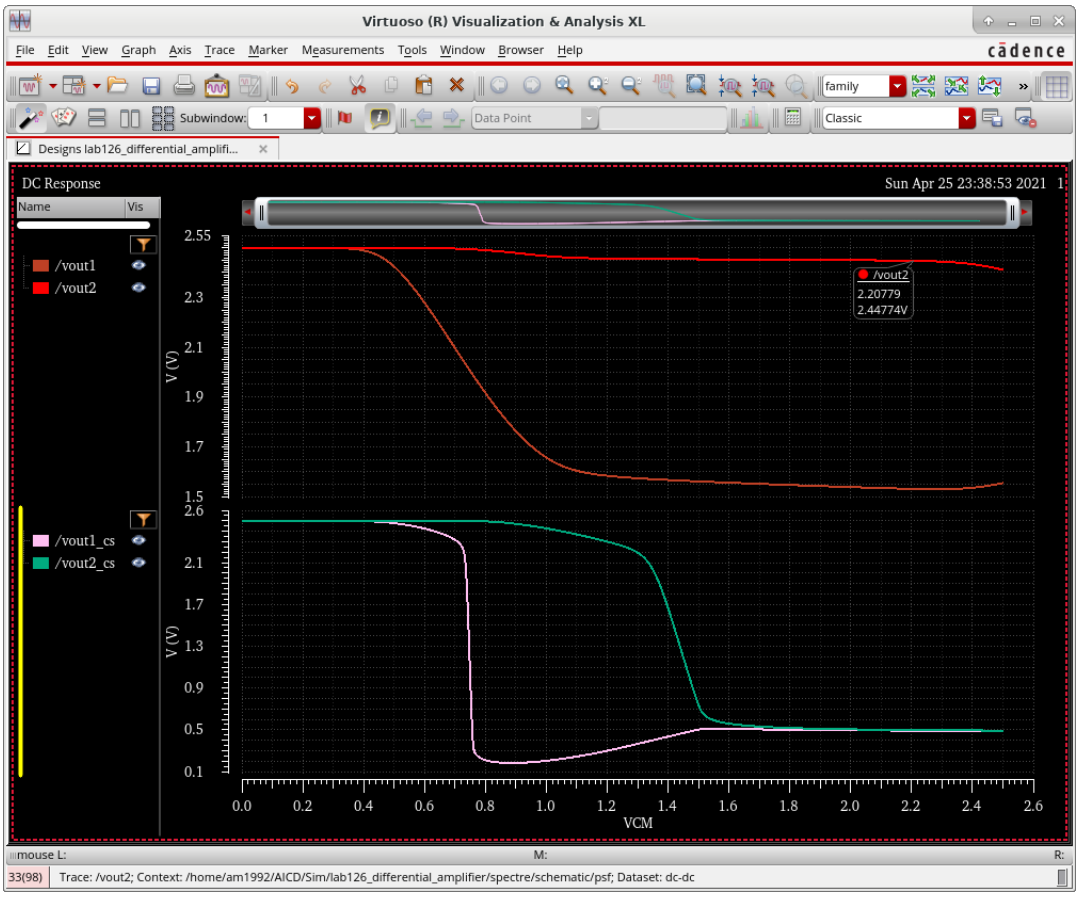
(5.2.9) Run the '**Netlist and Run**' command by clicking on the green play button. When the simulation has completed, separate the traces for the resistor-loaded and current source loaded differential amplifies. Figures 25 and 26.



< Figure 191. The Choosing Analyses Form. >



< Figure 192. The ADE with the Design Variables, Analyses and Outputs. >



< Figure 193. Resistor-Loaded and Current-Source Loaded Differential Amplifier Output Voltages. VD=250mV. VCM=0V to +2.5V >

(5.2.10) **Discussion**. Figure 26 plots the differential output voltages (vout1…, vout2…) of the differential amplifiers as the common-mode input voltage (VCM) is swept. We will use the waveform calculator to plot the common-mode output voltage (VCMO) for each of the amplifiers.

Note that the differential component of the output voltage rides on top of the common-mode component of the output voltage. We can eliminate the differential component by adding the two output voltages (*vout1…, vout2…*) and dividing by two. That is,

$$V_{CMO} = \frac{vout1 + vout2}{2}$$

In the calculator buffer, construct the expression for the common-mode output voltage (V_{CMO}): (VS("/vout1")+VS("/vout2"))/2. Do this for both the resistor-loaded and current source-loaded amplifiers. Add the VCMO plots to the respective VCM sweeps in the waveform window. See Figures 27 and 28.

We observe for the two differential amplifiers, there is a range of the VCM sweep over which the common-mode output voltage (yellow traces) remains approximately flat (at the DC operating point). This region for which there is no change in output voltage for a change in the common mode input VCM identifies the common mode input voltage range.

For the Resistor loaded differential amplifier (top) we see the common mode output voltage remains flat (at ~2V, the DC operating point) from around VCM=1.3V to VCM=2.5V. For the common source loaded amplifier we see the common mode output voltage remains flat (at ~1.3V, the DC operating point) from around VCM=0.8V to VCM=1.3V.

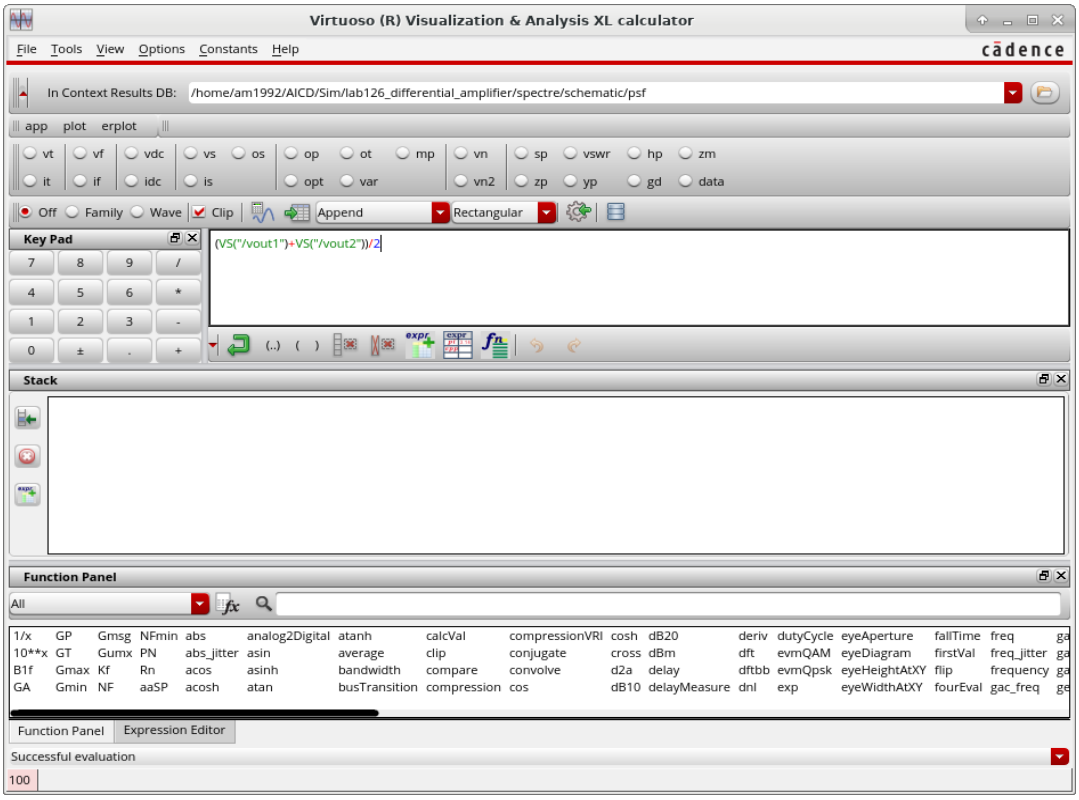
The principal question we want to answer is: what is the optimum value of common-mode input voltage (VCM) we should choose to bias the amplifiers?

This is the value of VCM that maintains the DC operating point for maximum swing (mid-rail, VDD/2) and maximises the differential gain.

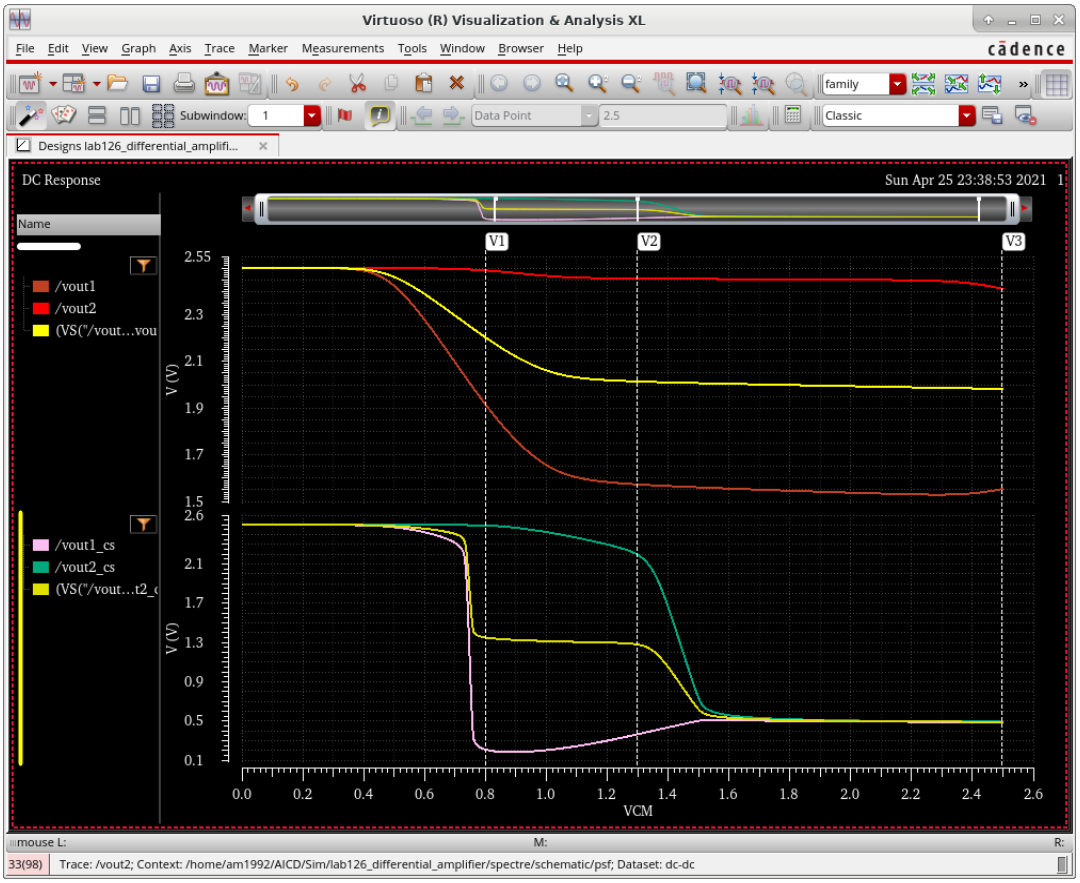
1) For the resistor-loaded amplifier, the combination of the linear load devices (resistors) and relatively low gain means biasing anywhere over the VCMO range (VCM=1.3V to VCM=2.5V) would be fine. However, we note that since we can't set the DC operating point independently of the gain, the DC operating point is fixed at around 2V. This limits the output signal swing (but since the gain is low, this may not be a problem).

2) For the current source loaded amplifier, the range is more limited, but we note that the differential gain is largest at around at VCM=0.9V. At VCM=0.9V the output voltage is very close to mid-rail at around 1.3V. So, the optimum choice for VCM for the current source loaded amplifier is 0.9V.

Make sure you understand how to define the common mode input voltage VCM(I) range, the common-mode output voltage VCM(O) range, and the factors that maximise/limit them!



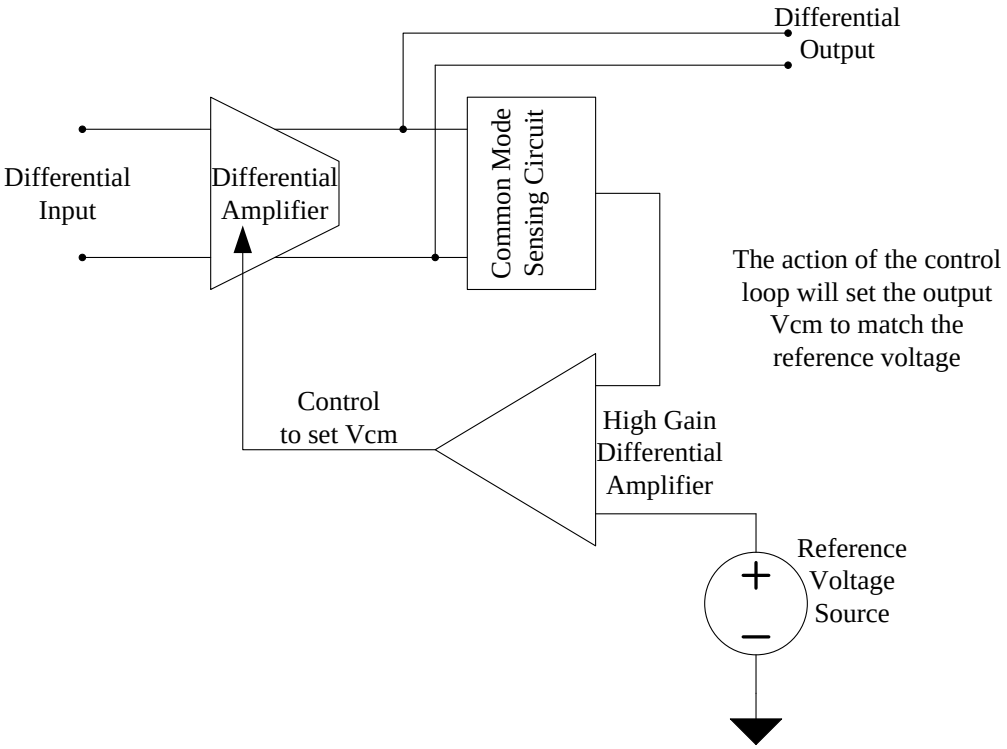
< Figure 194. Calculator Expression for Calculating Common-Mode Output Voltage. >



< Figure 195. Resistor-Loaded and Current-Source Loaded Differential Amplifier Output Voltages. With Calculator Plotted Common-Mode Voltages (yellow traces). VD=250mV. VCM=0V to +2.5V >

5.3 Common-Mode Feedback (CMFB)

In this part, of the lab we will provide a brief demonstration, using ideal components, of how we can use feedback to set the common-mode output voltage of the differential amplifier. This is a much more practical approach to setting the common-mode output voltage than we used in the previous part of this lab. It also makes the circuit robust to variations in process, voltage, and/or temperature (PVT). Figure 29 shows the basic configuration of the common-mode feedback system (CMFB).



< Figure 196. High-Level Diagram of the Common Mode Feedback System.

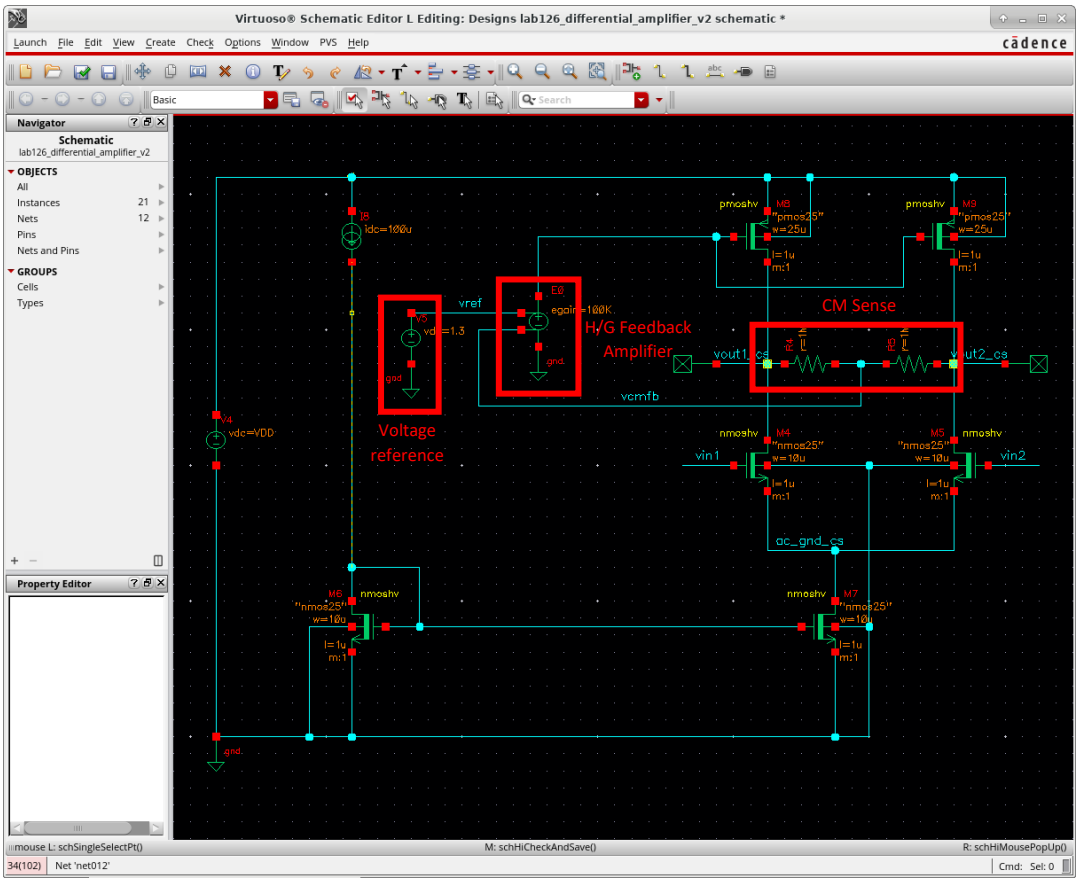
Reproduced from a figure used in a previous version of the Background Material. >

In Figure 29, the differential amplifier (top left) is the amplifier for which we want to set the common-mode output voltage. A sensing circuit (middle) is used to detect the common mode voltage from the outputs of the differential amplifier. The desired common-mode voltage is set by a reference voltage source (bottom right). A high gain feedback amplifier is used to compare the actual common mode output voltage with the desired common-mode output voltage set by the reference. The high gain amplifier is then used to drive the biasing circuitry of the differential amplifier so that the actual common mode output voltage is driven equal to the reference voltage.

(5.3.1) Create a new schematic of the fully differential amplifier. You can build one from scratch or you can copy your existing schematic using the copy command in the library manager. You may also modify your existing schematic by creating another circuit on the same canvas (alongside the two previous circuits you have already built – make sure to change the name of any named nets so as not to create an implicit connection to any of the other circuits on the same canvas).

(5.3.2) We will apply common-mode feedback to the current-source loaded differential amplifier we simulated in part 2 (Figure 13). Build the differential amplifier. Since we will use a feedback amplifier to bias the PMOS current source loads, do not include the PMOS mirror/ biasing circuitry we used in the last part.

(5.3.2) Instantiate and place two 1MΩ resistors (**res**), a Voltage-Controlled Voltage Source (**vcvs**) with a gain of 100K, a voltage reference set to 1.3V (**vdc**), and a ground connection (**gnd**). Source all components from the analogLib ideal component library. See Figure 30.



< Figure 197. The Current-Source Loaded Differential Amplifier Modified for Common-Mode Feedback. >

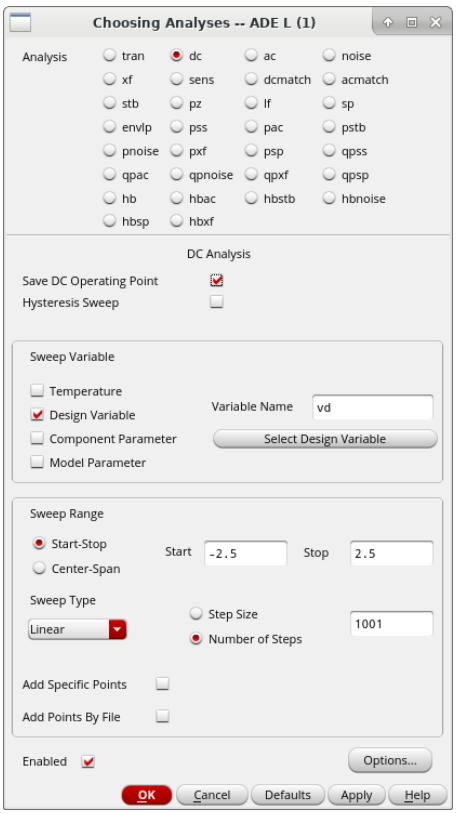
(5.3.3) Note: Here we are using the two 1MΩ resistors as a basic common-mode sense circuit. Since the output voltages are complementary (one goes up, one goes down), the mid-point between the two resistors gives the common voltage between the two outputs. We are using two very large resistors so as not to load the amplifier and reduce its gain. This is not a practical solution – practical designs would use a transistorised circuit design to sense the common mode voltage. However, this setup is useful to demonstrate the principle of common-mode feedback.

(5.3.4) We chose a value of 1.3V for the common-mode output voltage – set by the reference **vdc**. This is just above the mid-rail. Note that the VCM is set at 1.25V, and we want drain voltage of the input pair to be higher than the gate voltage of the input pair to ensure they remain in saturation.

(5.3.5) For the feedback amplifier (**vcvs**) we use a high gain of 100K to ensure high tracking accuracy between the actual output and the reference voltage. **It is important to note that we are applying the feedback to the PMOS current source load devices. It is through this feedback that we are matching the PMOS and NMOS currents for a given VCMO.**

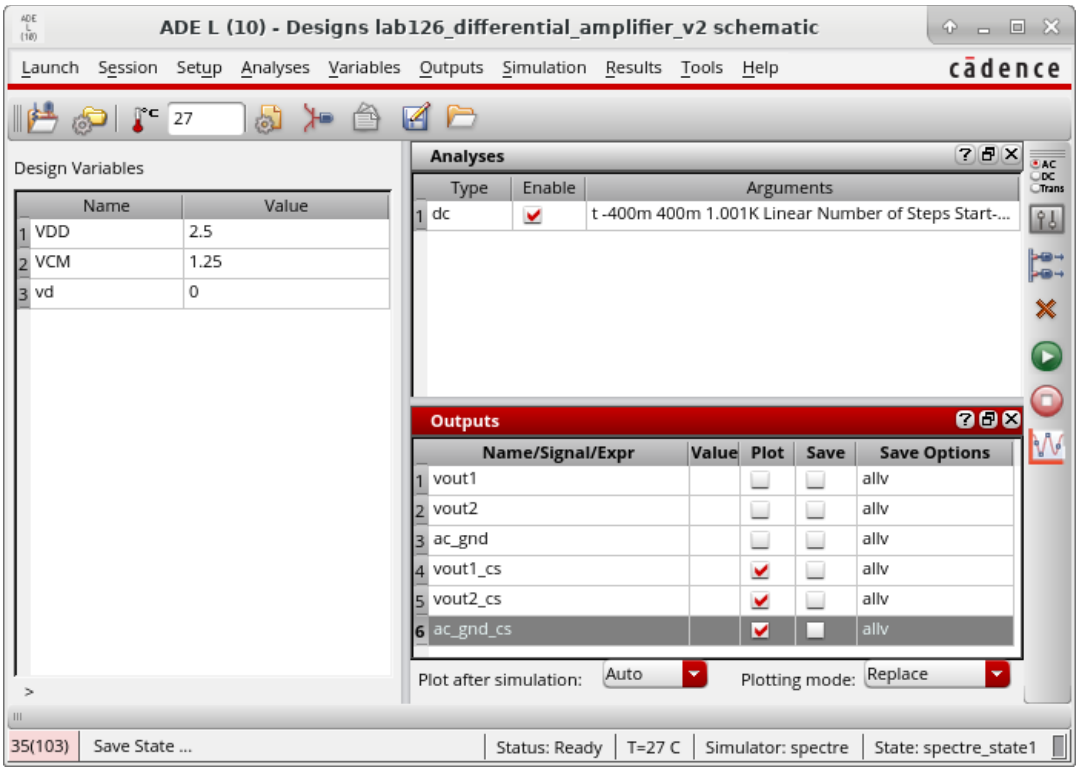
(5.3.6) Launch the ADE. As before, we want to plot the differential output voltage (vout1_cs and vout2_cs) and the AC ground point (ac_gnd_cs) as we sweep v_D from -2.5 to 2.5V. From the ADE window, go to **Menu: Analyses à Choose…** or use the '**Choose Analyses**' button on the right-side of the ADE window (top button). The 'Choosing Analyses – ADE L' form will open. From the form, select the 'dc' option button. Check the 'Save DC Operating Point'.

In the choosing analyses form. Check the 'Save DC Operating Point' box. Check the 'Design Variable' box as the sweep variable. Type vd as the variable name. Choose the 'Start-Stop' option button with Start = -2.5 and Stop = 2.5 for the sweep range. Select 'Linear' from the drop-down menu as the sweep type. Choose the 'Number of Steps' option button with 1001 steps in the sweep. Ensure that the analysis is enabled by checking the 'Enabled' box near the bottom of the form. See Figure 6. OK the form. The **dc** analysis should now appear in the 'Analyses' list of the ADE window.

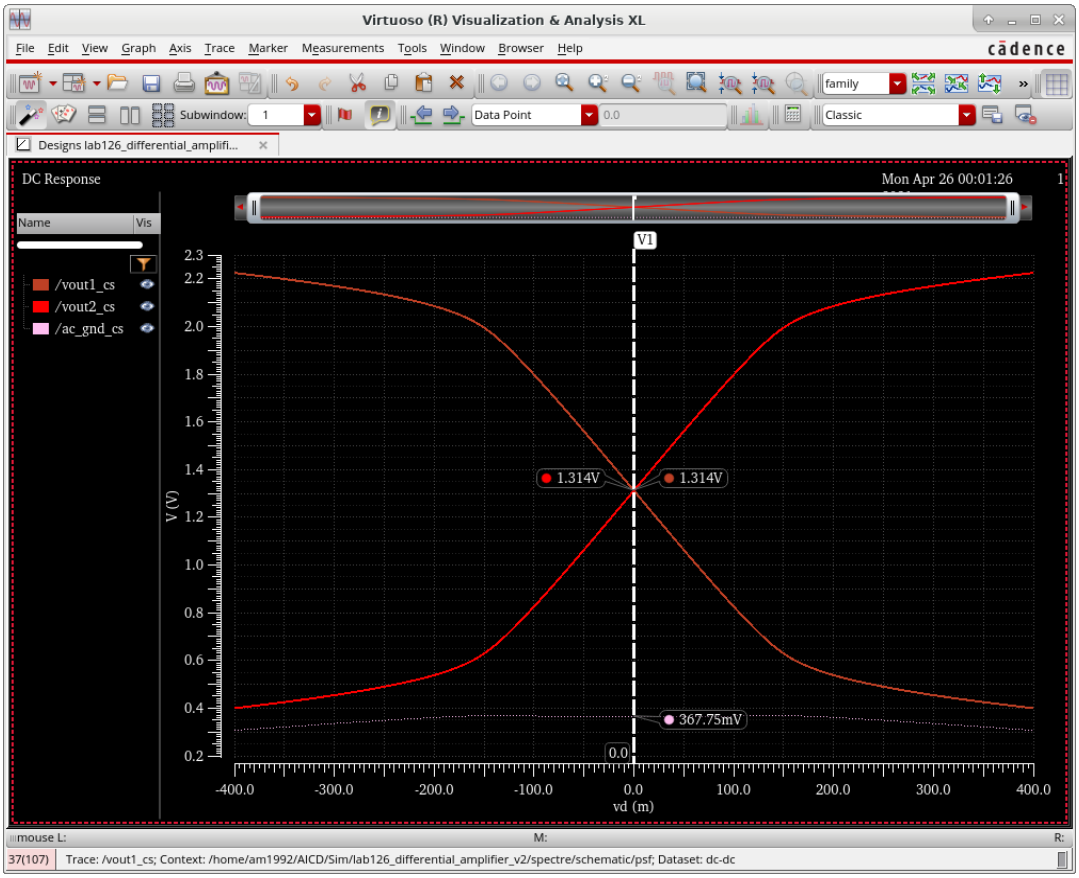


< Figure 198. The Choosing Analyses Form. >

(5.3.7) Select the two output voltages (vout1_cs and vout2_cs) and the AC ground point (ac_gnd) for plotting. In the Analog Design Environment window, select **Menu: Outputs à To Be Plotted à Select On Schematic** to enter the plotting mode of the ADE**. Press the 'ESC' key to exit the plotting mode of the ADE**.



< Figure 199. The ADE with the Design Variables, Analyses and Outputs. >



< Figure 200. Current-Source Loaded Differential Amplifier Output Voltages Set Using Common-Mode Feedback. VCM=1.25V. VD=-0.4V to +0.4V >

(5.3.8) **Discussion**. Referring to Figure 33. Using the common-mode feedback setup we have set the common-mode output voltage to 1.314V (our reference was 1.3V). Note, that in practice we do not need such a high gain (100K) to get a good tracking accuracy. For this simple demonstration, the **VCVS** is also providing the DC biasing for the circuit which is problematic since its gain function cannot be decoupled from its DC biasing function. In a practical feedback circuit, this would may be an issue.

You are encouraged re-simulate the feedback circuit for different values of reference voltage and VCVS gain.

End of Lab!